An Illustrated Course in Real Analysis

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Welcome

Real analysis occupies a special place in the psyche of undergraduate math majors and instructors. The course is widely regarded as the most difficult required course for the major. And yet for many, it fundamentally changes their relationship with mathematics. Getting through the course is a challenge, but the subject's simultaneous applicability and dedication to rigor often resonates with both learners and teachers.

I think for this reason, teaching the course is very personal and each instructor wants to bring their own flavor of understanding into the classroom. For me, illustrating and visualizing each concept is an important component of understanding, and this informs my take on the course in this text.

This version of the textbook is still in flux, with updates and fixes constantly being made. I hope to eventually have a stable "first edition".

Book Features and Structure

Each chapter is meant to correspond to one day of instruction. When this disrupted the natural grouping of topics and theorems, I did my best. The number of chapters corresponds to a typical number of days of primary instruction for a three-times-per-week, 14-week course (leaving days for review, exams, etc).

Desmos Interactives

The most distinguishing feature of this textbook is the incorporation of interactive illustrations created with Desmos. These were created using the Desmos API and are included in this textbook with permission from Desmos Studio PBC.

Each illustration was carefully designed with the learner in mind. While I have a great deal of fun creating these things, I tried to exercise restraint. I feel strongly that students should get in the habit of drawing their own pictures, and many of the exercises explicitly ask them to do so.

Most of the illustrations in the textbook have the equations and controls hidden. Links to versions that can be seen and edited on the Desmos web app are available in Appendix E.

Annotated Proofs

Many of my proofs are annotated with mouse-overs.

It is no secret that the students who struggle most in undergraduate real analysis are ones who don't enter the course with solid basic proof-writing skills. And when dealing with complicated stacked quantifiers (as in the definition of continuity), bright students who had gotten by on intuition alone up until this course often find themselves struggling and need to return to the basics.

For these reasons, I include some highly systematic proofs that go "by the book" in terms of dealing with quantifiers and using standard proof structures, with annotations explaining every step.

$f(x) = 3x + 1$ *is continuous at* $a = 4$. We need to prove that

$$(\forall \epsilon > 0)\ (\exists \delta > 0)\ (|x - 4| < \delta \implies |f(x) - f(4)| < \epsilon)$$

Suppose $\epsilon > 0$. Let $\delta = \frac{\epsilon}{3}$. Suppose $|x - 4| < \delta$ Then

$$\begin{aligned} |f(x) - f(4)| &= |(3x + 1) - (3 \cdot 4 + 1)| \\ &= |3(x - 4)| \\ &\vdots \end{aligned}$$

□

The resulting proofs are very dry, and I don't necessarily recommend doing every proof so systematically, but I hope that it gives students something to fall back on.

Structure of Each Chapter

Anyone should feel free to use this textbook however they'd like, but here were my intentions while writing.

- *Highlights* give a quick overview of what's covered. I think this is especially helpful for the web rendering.
- The subsection *Preparation* is intended as pre-class readings and exercises for students. I think it's important for math majors to get practice *actively reading mathematics*. Designating the introduction to each section as the required reading is a sort of compromise, and they are meant to be especially student-readable.
- Subsequent sections outline what might be covered during class. Many proofs, even of major theorems, are left as exercises. This gives the instructor the option of assigning them to students (for homework or during class).
- *Review Questions* at the end of each section are meant to be useful for students to go through right after class or while studying for an exam.

Notes on the content

This textbook was written with the course *MA 425: Mathematical Analysis I* at North Carolina State University in mind. Officially, the course description reads:

> Real number system, functions and limits, topology on the real line, continuity, differential and integral calculus for functions of one variable. Infinite series, uniform convergence.

Here are a few decisions I made in curating the content.

- Even though it isn't required, I spend a bit of time on metric spaces in general. This is because I believe we gain a lot of intuition about distances by thinking about geometry in $\mathbb{R}^2$ and $\mathbb{R}^3$; for some reason geometry is actually harder for our brains in $\mathbb{R}^1$. It's also often the case that proving something for a general metric space is no harder than proving it for $\mathbb{R}$; and knowing that the proof should be general lets students focus on using metric space properties without getting distracted by $\mathbb{R}$'s additional structure. The philosophy in general is "get intuition from $\mathbb{R}^2$ and $\mathbb{R}^3$; prove for general metric spaces; apply in $\mathbb{R}$".
- While dealing with general metric spaces in principle could allow us to talk about the topological definition of compactness, this topic is relegated to an appendix; one needs to pick one's battles.

Technology Used in Authorship

This textbook was created using Quarto, which is designed for multiformat output. A PDF version is available for download, but of course, the interactivity is lost.

As stated above, the Desmos API is used in this textbook with permission from Desmos Studio PBC.

I used OpenAI tools (Codex and ChatGPT) to assist with technical aspects of the textbook including formatting, troubleshooting interactivity for the Desmos interactives and annotated proofs, and proof-reading.

Every word of the written content—typos, errors, hallucinations, goblins, em-dashes, and all—is my own.

Licensing

This work is licensed under CC BY-NC-SA.

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I also want to acknowledge and thank NC State University and the Mathematics Department in particular for granting my sabbatical to participate in the program and for travel support.

The Desmos API is used with Permission from Desmos Studio PBC.

Thank you to the Desmos Studio Public Benefit Corporation for allowing me to use your wonderful product in this text.

I also acknowledge additional support from an NC State University Libraries Alt-Textbook Project grant, and thank David Tully and Will Cross for their support and advice.

Part of the organization and presentation are inspired by *Closer and Closer* by Carol Schumacher (2008) and *Understanding Real Analysis* by Stephen Abbott (2016).

Finally, I want to acknowledge John Rock, author of *Arbitrarily Close* (2025), another real analysis textbook with dozens of wonderful linked Desmos interactives. John has been extremely generous in sharing his experiences in authoring in this format.

1 Introduction

Coming soon.

Part I

The Real Numbers

2 The Field and Order Axioms

Highlights

- Two of the three axioms we will assume about the real number system: the field axiom which gives us the rules of arithmetic, and the order axiom which gives us the rules for inequalities.
- We see examples of other number systems besides $\mathbb{R}$ that also satisfy one or both of these axioms. In particular $\mathbb{Q}$ also satisfies both.

2.1 Preparation

As discussed in the introduction, our approach to real analysis will be to start by laying out a set of axioms that we *assume* are true about the real number system $\mathbb{R}$ and then *prove* everything else based upon this set of axioms.

In order to properly take this approach, we need a set of axioms that:

- contains enough information to let us prove everything there is to know about $\mathbb{R}$ (all of calculus, for example),
- is specific enough so that it distinguishes $\mathbb{R}$ from other number systems, and
- is as simple as possible

The purpose of the first bullet point is clear enough: if I don't assume anything, I won't be able to prove anything.

The second bullet point about distinguishing $\mathbb{R}$ from other number systems is related, but deserves some explanation. Consider the following proposition:

> For every $x > 0$, there exists y such that $y^2 = x$.

This is true if our universe is $\mathbb{R}$ but false our universe is $\mathbb{Q}$. If our axioms for $\mathbb{R}$ do not distinguish it from $\mathbb{Q}$, we will have no hope of proving such a proposition. In short, I want a set of axioms that is true for $\mathbb{R}$ *but for no other number system.*

> **Note**
>
> I am using the term *number system* a bit loosely. For our purposes, though, it's fine to think of any set with binary operations corresponding to addition and multiplication—think $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$.

Look ahead to the definition of a field (Definition 2.2) and definition of an ordered field (Definition 2.7). See if you can convince yourself the real number system does indeed satisfy both of these definitions. For example, every real number x *does* in fact have an additive inverse (a number $-x$ that be added to x to obtain 0); it *is* true that when you multiply two positive real numbers you get another positive real number.

Take another moment to scan some of the *theorems* that follow each of these axioms. None of these should be surprising—they are all well-known facts about how arithmetic and inequalities work in $\mathbb{R}$—but what may be surprising is that extremely simple facts like $0x = 0$ (Theorem 2.4) and $1 > 0$ (Theorem 2.9) can be *proven* from even simpler assumptions.

For now, let's focus on thinking about which other familiar number systems satisfy the field and order axioms.

Exercise 2.1. Think about some familiar number systems that *aren't* the real numbers: $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, and $\mathbb{C}$. You may rely on prior knowledge for this problem—no need to show all of your work.

1. Which of these are fields? For those that are not, which properties do they fail to have?
2. Which of these are ordered fields? Which of these satisfy the three conditions of Definition 2.7 despite not actually being a field?
3. **(Exploratory)** The Intermediate Value Theorem is true in $\mathbb{R}$, but it is *not* true in any of the other four number systems from this problem. You should have found that one of the four number systems is a field and an ordered field. With that information, do you think one would be able to prove the Intermediate Value Theorem in $\mathbb{R}$ if our *only* assumptions about $\mathbb{R}$ are the field and order axioms?

2.2 Fields

For our purposes, a **number system** $\mathbb{S}$ is a set with two binary operations $+$ and $\cdot$. This is a nonstandard term, but we won't use it much beyond this section.

Definition 2.2. A number system $\mathbb{S}$ is said to be a **field** if it satisfies the following properties.

1. Addition is commutative (for all $a, b \in \mathbb{S}$, $a + b = b + a$).
2. Addition is associative (for all $a, b, c \in \mathbb{S}$, $(a + b) + c = a + (b + c)$).

3. There is an element $0 \in \mathbb{S}$, called the *additive identity* such that for all $a \in \mathbb{S}$, $a + 0 = a$.
4. Every $a \in \mathbb{S}$ has an *additive inverse* $-a$ with the property that $a + (-a) = 0$.
5. Multiplication is commutative (for all $a, b \in \mathbb{S}$, $a \cdot b = b \cdot a$).
6. Multiplication is associative (for all $a, b, c \in \mathbb{S}$, $(a \cdot b) \cdot c = a \cdot (b \cdot c)$).
7. There is an element $1 \in \mathbb{S}$ different from 0, called the *multiplicative identity* such that for all $a \in \mathbb{S}$, $a \cdot 1 = a$.
8. Every $a \in \mathbb{S} \setminus \{0\}$ has an *multiplicative inverse* a^{-1} with the property that $a \cdot (a^{-1}) = 1$.
9. The *distributive rule* holds: for all $a, b, c \in \mathbb{S}$, $a(b + c) = a \cdot b + a \cdot c$.

Collectively, these properties tell us how **arithmetic** works for fields. And the first axiom we make about the real number system is that it is indeed a field.

Axiom 2.3. $\mathbb{R}$ *is a field.*

Note that there is no reference to the operations of subtraction and division in the definition of a field, but these can be *defined* in terms of the established properties: $a - b$ is defined to be $a + (-b)$ (after all, $-b$ is defined above), and if $b \neq 0$, we define $\frac{a}{b}$ to be $a \cdot b^{-1}$ (after all, b^{-1} is defined above).

What is somewhat remarkable is that *all* of the other rules of arithmetic for the real numbers can be derived from the field axiom. For example, you may have supposed that the following theorem would be an assumption, but one can prove it from the even more basic assumptions in the definition of a field.

Theorem 2.4. *For any* $x \in \mathbb{R}$, $0x = 0$.

Exercise 2.5. Prove Theorem 2.4.

> Hint
>
> Note that Theorem 2.4 involves both multiplication and, implicitly, addition (since it involves the *additive* identity). Thus its proof must involve the one part of the field axiom that combines addition and multiplication: the distributive property.

Any facts that we can prove about $\mathbb{R}$ that use *only* the assumption that $\mathbb{R}$ is a field, will in fact be true for *any* field. And there are some weird fields.

Exercise 2.6. Consider the number system $\mathbb{F}_2 = \{\alpha, \beta\}$ consisting of just two elements, where the addition and multiplication operations are defined by

$$\alpha + \alpha = \beta \qquad \alpha + \beta = \alpha \qquad \beta + \alpha = \alpha \qquad \beta + \beta = \beta$$

and

$$\alpha \cdot \alpha = \alpha \qquad \alpha \cdot \beta = \beta \qquad \beta \cdot \alpha = \beta \qquad \beta \cdot \beta = \beta.$$

It turns out that $\mathbb{F}_2$ is a field. Which element of $\mathbb{F}_2$ is 0? Which is 1? Which is -1?

> **Hint**
>
> Asking which element is 0 means asking which element is the additive identity—the thing that can be added to any element and get that same element back. Similarly for the multiplicative identity 1. And asking which element is -1 means asking which element is *the additive inverse of* 1 (the thing you add to 1 and get back 0).

So the field axiom alone fails miserably at the goal of distinguishing $\mathbb{R}$ from other number systems—it cannot even distinguish $\mathbb{R}$ from a number system with just two elements! So we need more axioms capturing more of the structure of $\mathbb{R}$.

2.3 Order and Inequalities

Another important feature of the real number system is the behavior of inequalities and how they interact with arithmetic. This is captured by the next axiom.

Definition 2.7. A field $\mathbb{F}$ is called an **ordered field** if it has a relation $>$ with the following properties.

1. For all $a \in \mathbb{F}$, exactly one of the following is true:
   - $a > 0$,
   - $a = 0$, or
   - $-a > 0$.
2. If $a > 0$ and $b > 0$, then $a + b > 0$ and $a \cdot b > 0$.
3. $b > a$ if and only if $b - a > 0$.

We can then *define* relations $<, \leq, \geq$ in the usual way based on the relation $>$[1].

Axiom 2.8. *$\mathbb{R}$ is an ordered field.*

Again, the properties listed in Definition 2.7 are not surprising; we already knew they were true for $\mathbb{R}$. But more surprising is the fact the everything else we know about how inequalities behave and interact with arithmetic can be prove from Axiom 2.3 and Axiom 2.8.

[1]That is, we define $a < b$ to mean $b > a$, $a \leq b$ to mean $a < b$ or $a = b$, and $a \geq b$ to mean $a > b$ or $a = b$.

Theorem 2.9. $1 > 0$.

Exercise 2.10. Prove Theorem 2.9 using only the Field and Order Axioms.

You are probably familiar with the basics of how inequalities and arithmetic interact, and henceforth, we can use them at will without citation. In this class, being fluent with the rules for inequalities will be essential, though, so we collect a list of "inequality rules" (all of which could in principle be proven from Axiom 2.3 and Axiom 2.8) in Section 2.4.

Worth reflecting on now is where we stand on our stated goals in creating the axioms. The Axiom 2.8 was successful in distinguishing $\mathbb{R}$ from $\mathbb{F}_2$ because there is no ordering on $\mathbb{F}_2$ satisfying Definition 2.7 (you will prove this in Exercise 2.11 below).

However, we can note that $\mathbb{Q}$ is also an ordered field since it satisfies Axiom 2.8. This means that anything we can prove directly from Axiom 2.3 and Axiom 2.8 will apply just as well if our universe were $\mathbb{Q}$ as it would if our universe were $\mathbb{R}$. In particular, we would have no hope of proving, say, the Intermediate Value Theorem, which is true in $\mathbb{R}$ but not in $\mathbb{Q}$.

This all means that we need another assumption—an axiom that will distinguish the real number system from the rationals in a way that will allow us to prove everything else we'll need in order to do calculus.

Exercise 2.11. Prove that the following fields are *not* ordered.

1. $\mathbb{F}_2$ (defined in Exercise 2.6)
2. The complex numbers $\mathbb{C}$

> **Hint**
>
> This is a good opportunity to use a proof by contradiction. What contradiction can you create by assuming there *is* an ordering on $\mathbb{F}_2$ satisfying the conditions of Definition 2.7? What about in $\mathbb{C}$?

The following exercise says something very simple, but ends up being very useful in practice.

Exercise 2.12. Suppose $r \geq 0$ and that for all $\epsilon > 0$, $r \leq \epsilon$. Prove that $r = 0$.

> **Note**
>
> Now that you've finished this section, you can use whatever facts you want about arithmetic and order in $\mathbb{R}$ without citing theorems or axioms.

2.3.1 Review Questions

1. Give some examples of number systems that are fields. Give some examples of number systems that are not fields.
2. Give some examples of fields that are ordered fields. Give some examples of fields that are not ordered fields.
3. Summarize for yourself how we can be certain that it is impossible to prove a statement like "every positive number in $\mathbb{R}$ has a square root" building off of only the field and order axioms.

2.4 Inequality Rules

- a is positive $\Leftrightarrow -a$ is negative
- $1 > 0$
- $x \leq y$ and $y \leq z \Rightarrow x \leq z$
- $a > b$ and $c \geq d \Rightarrow a + c > b + d$
- $a > b > 0$ and $c \geq d > 0 \Rightarrow ac > bd$
- $a > b$ and $c < 0 \Rightarrow ac < bc$
- $a < 0$ and $b < 0 \Rightarrow a + b < 0$
- $a < 0$ and $b < 0 \Rightarrow ab > 0$
- $a > 0$ and $b < 0 \Rightarrow ab < 0$
- $a > 0 \Leftrightarrow \frac{1}{a} > 0$
- $a > 1 \Rightarrow a^2 > a$
- $0 < a < 1 \Rightarrow a^2 < a$
- $a > b > 0 \Rightarrow a^2 > b^2$

3 The Least Upper Bound Axiom

Highlights

- The definition of the supremum of a set and the Least Upper Bound axiom of $\mathbb{R}$
- The Archimedean Properties of $\mathbb{R}$

3.1 Preparation

In this section, we will lay out an axiom for $\mathbb{R}$ that distinguishes it from other ordered fields. Leading up to this axiom, we make a few definitions.

Definition 3.1. Let $S \subseteq \mathbb{R}$ be a subset of the real numbers.

1. We say $b \in \mathbb{R}$ is an **upper bound** of S if for all $x \in S$, $b \geq x$. If S has an upper bound, we say S is **bounded above**.
2. We say that $u \in \mathbb{R}$ the **least upper bound** or **supremum** of S if u is an upper bound and for all upper bounds b of S, $u \leq b$. In this case we write
$$u = \sup S.$$
3. If u is the least upper bound S and $u \in S$, we say that u is the **greatest element** of S.

Exercise 3.2.

1. Write down three upper bounds for the interval $S = [0, 3)$. Does S have a least upper bound; if so what is it? Does S have a greatest element; if so, what is it?
2. For each set below, say what it's supremum and greatest element are or say why it doesn't have one.
   a. $\left\{\frac{n}{n+1} \ : \ n \in \mathbb{N}\right\}$
   b. $(2, \infty)$
   c. $\left\{\frac{1}{n} \ : \ n \in \mathbb{N}\right\}$
   d. $(-\infty, 3]$
   e. $\{x \ : \ x^2 \leq 3\}$

3. Describe in your own words which subsets of $\mathbb{R}$ have upper bounds, which have least upper bounds, and which have greatest elements. This doesn't have to be precise; just give me the vibe.
4. **(Exploratory)** How would you change your answers to the previous problems if our universe were $\mathbb{Q}$ instead of $\mathbb{R}$?

Exercise 3.3. Write "S is bounded above" in logical symbolic notation; see if you can do so without using the term "upper bound".

Exercise 3.4. Suppose b is an upper bound on a set S. What does it mean for b to *not* be the *least* upper bound of S?

3.2 Least Upper Bounds

Part 2e of Exercise 3.2 is meant to draw a contrast between how least upper bounds work in $\mathbb{R}$ versus how they work in $\mathbb{Q}$. In particular, the problem asks, if

$$S = \{x \ : \ x^2 \leq 3\}$$

does S have a least upper bound?

Whether our universe is $\mathbb{Q}$ or $\mathbb{R}$ or some other ordered field, S certainly has *an* upper bound; $b = 2$, for example, is greater than any element of S (that is, 2 is greater than any number whose square is less than 3). But as you may have guessed, an ordered field has a number whose square is 3 if and only if the set S defined above has a *least* upper bound (in which case $\sup S$ is such a number).

We have a sense that the since 3 has no square root in $\mathbb{Q}$, there is something missing—a "hole" in the rationals that we plug by instead thinking about $\mathbb{R}$. And as this example shows, plugging such a hole can be related to the existence of a least upper bound on a set.

This connection goes in the opposite direction as well. If we *assume* the supremum of S exists, then we at least have a candidate for a number whose square is 3 (actually proving this is more complicated than you might expect—see Theorem 3.16 below).

This is all motivation for the final axiom we'll assume about the real numbers: a guarantee of the existence of least upper bounds that will have the effect of ensuring that any $\mathbb{R}$ will not have any "holes" in this sense, the way that $\mathbb{Q}$ does.

Axiom 3.5 (The Least Upper Bound Axiom)**.** *Every non-empty subset of* $\mathbb{R}$ *that is bounded above has a least upper bound in* $\mathbb{R}$.

Tip

Note that using Axiom 3.5 to conclude that a subset $S \subseteq \mathbb{R}$ has a supremum requires S to satisfy two hypotheses:

- S is nonempty, and
- S is bounded above.

From now on, we'll make a point of emphasizing which theorems require the Least Upper Bound Axiom in their proofs and how. Theorems that rely on the Least Upper Bound Axiom will be highlighted with a maroon bar.

As a first step to understanding least upper bounds, we should justify the use of the word "the" in Definition 3.1.

Tip

Generally, when define a new quantity implicitly (i.e. without giving an *ex*plicit formula), we should ask ourselves whether a *unique* quantity satisfies the definition. For example, a subset of $\mathbb{R}$ may have more than one upper bound. That a subset has only one *least* upper bound (if it has one at all) is something we need to prove.

Theorem 3.6. *If a subset of $\mathbb{R}$ has a least upper bound, then the least upper bound is unique.*

Proof. Suppose u, v are both least upper bounds for a subset $S \subseteq \mathbb{R}$. Then u is an upper bound for S, so since v is a *least* upper bound, $v \leq u$. Also, v is an upper bound for S, so since u is the *least* upper bound, $u \leq v$. Therefore $u = v$. □

To prove a quantity is the least upper bound of a set, generally, one needs to show two things: prove that it's an upper bound, and then prove that any other upper bound is less than or equal to it.

Example 3.7. Here we'll do a careful proof that 3 is the supremum of $S = (-\infty, 3)$.

First, note that by definition S is all numbers less than 3, so for any $x \in S$, $3 > x$. Therefore 3 is an upper bound on S.

By way of contradiction, suppose $b < 3$ and b is also an upper bound on S. But then consider $x = \frac{b+3}{2}$ so that $b < x < 3$. Since $x < 3$, we know $x \in S$. But $b < x$ which contradicts b being an upper bound on S.

Note that doing a proof by contradiction above was not really necessary. Another option would be to say "suppose $b < 3$" and then prove b is not an upper bound.

Next we have some basic consequences of Axiom 3.5. The next theorem and its corollary are both often referred to as the "Archimedean Property" of $\mathbb{R}$.

Theorem 3.8 (Archimedean Property - Version 1)**.** *For any $x \in \mathbb{R}$, there exists $n \in \mathbb{N}$ such that $n > x$.*

Proof. By way of contradiction, suppose _____, which is the same as saying that x is an upper bound on $\mathbb{N}$. Then since $\mathbb{N}$ is _____ and bounded above, $\mathbb{N}$ has a least upper bound u. But then $u - \frac{1}{2} < u$, so $u - \frac{1}{2}$ is not an upper bound. Therefore there exists $n \in \mathbb{N}$ such that _____. But then

$$n + 1 > u - \frac{1}{2} + 1 = u + \frac{1}{2} > u,$$

and since $n + 1 \in \mathbb{N}$, we have contradicted _____. □

Exercise 3.9. Fill in the blanks for the proof above. For the first blank, you'll need to negate the original statement.

Corollary 3.10 (Archimedean Property - Version 2)**.** *For any $\epsilon > 0$, there exists $n \in \mathbb{N}$ such that $\frac{1}{n} < \epsilon$.*

Exercise 3.11. Prove Corollary 3.10.

> **Hint**
>
> Remember that when we call a theorem a "Corollary", it means that you should use the preceding theorem to prove it, and the proof will generally end up being quite short.

Exercise 3.12. Give a careful proof that 1 is the supremum of $S = \left\{ \frac{n}{n+1} \ : \ n \in \mathbb{N} \right\}$.

> **Hint**
>
> The same trick of taking an average like in Example 3.7 won't necessarily work since that average might not be in S. Corollary 3.10 might be helpful here instead.

Exercise 3.13. Suppose $S \subseteq \mathbb{R}$ is a nonempty set that is bounded above. Let $c > 0$. We define the set

$$T = \{cx \ : \ x \in S\}.$$

Prove that $c \cdot \sup S$ is a least upper bound for T.

Exercise 3.14. Suppose S and T are nonempty subsets of $\mathbb{R}$ that are bounded above. What is $\sup(S \cup T)$? Prove your answer.

Exercise 3.15. Prove the Archimedean Property is true when the universe is $\mathbb{Q}$. This means you'll need to refrain from making any references to $\mathbb{R}$, and you of course can't use the Least Upper Bound Axiom.

In Part 2e of Exercise 3.2 and the follow-up discussion, we claimed that the existence of suprema would "plug holes" in the real numbers. In particular, the supremum of $S = \{x \ : \ x^2 \leq 3\}$, which we are assuming exists because of the Least Upper Bound Axiom, seems like it ought to be a number whose square is 3. We'll prove a more general version of this fact below, although the proof gets somewhat technical towards the end.

Theorem 3.16. *For any $a \geq 0$, there exists a real number $x \geq 0$ such that $x^2 = a$.*

Proof. Since the case where $a = 0$ is trivial (since $0^2 = 0$),suppose $a > 0$. Define

$$S = \{y \ : \ y^2 \leq a\}$$

We will show that S is nonempty and bounded above so that we can claim $\sup S$ exists.

First, we note that $0^2 = 0 < a$, so $0 \in S$, and thus S is nonempty.

Next, I claim that $a + 1$ is an upper bound on S. This is true because

$$(a+1)^2 = a^2 + 2a + 1 > a$$

So if $y \in S$,

$$y^2 \leq a < (a+1)^2$$

so $y < a+1$. Thus $a+1$ is an upper bound on S, so S is bounded above, and $x = \sup S$ exists, and since $0 \in S$, $x \geq 0$.

By way of contradiction, suppose $x^2 \neq a$. There are two cases: $x^2 < a$ and $x^2 > a$, and we will finish the proof by showing each leads to a contradiction.

(Case 1: $x^2 < a$) Let $\epsilon = \min\left\{1, \frac{a-x^2}{2x+1}\right\}$ so that $\epsilon > 0$ and $\epsilon^2 \leq \epsilon$. Then

$$\begin{aligned}(x+\epsilon)^2 &= x^2 + 2x\epsilon + \epsilon^2 \\ &\leq x^2 + 2x\epsilon + \epsilon \\ &= x^2 + \epsilon(2x+1) \\ &\leq x^2 + a - x^2 \\ &= a.\end{aligned}$$

Then $x+\epsilon > x$ but $(x+\epsilon)^2 \leq a$, so $x+\epsilon \in S$. Thus x is not an upper bound on S, which is a contradiction.

(Case 2: $x^2 > a$) Here we know $x > 0$, so let $\epsilon = \min\left\{x, \frac{x^2-a}{2x}\right\}$ so that $\epsilon > 0$ and $x - \epsilon > 0$. Then

$$\begin{aligned}(x-\epsilon)^2 &= x^2 - 2x\epsilon + \epsilon^2 \\ &> x^2 - 2x\epsilon \\ &\geq x^2 - 2x\frac{x^2-a}{2x} \\ &= x^2 - (x^2 - a) \\ &= a.\end{aligned}$$

But since $(x-\epsilon)^2 > a$, this means that for any $y \in S$, $y^2 < a < (x-\epsilon)^2$, so

$$y \leq |y| < |x-\epsilon| = x - \epsilon,$$

so $x - \epsilon$ is an upper bound on S less than x, contradiction x being the *least* upper bound.

Thus, since $x^2 \neq a$ led to a contradiction, we must have $x^2 = a$. □

3.3 Review Questions

1. Give definitions of upper bound, least upper bound, and greatest element.
2. What does the Least Upper Bound Axiom say? What two things do we need to know about a set S in order to apply the Least Upper Bound Axiom?
3. For examples like those in Part 2 of Exercise 3.2, can you quickly identify the least upper bound when there is one?
4. What do the two versions of the Archimedean Property tell us?

4 Consequences of the Least Upper Bound Axiom

Highlights

- We define the greatest lower bound (also called the infimum) of a set and prove that these exist in $\mathbb{R}$ whenever a nonempty subset is bounded below.
- We prove the Nested Interval Theorem, which shows that the intersection of a nested sequence of closed intervals is nonempty.
- We show how the Nested Interval Theorem can be used to prove that $\mathbb{R}$ is uncountable.

4.1 Preparation

We start with a definition that should seem very natural given what we did in the last section.

Definition 4.1. Let $S \subseteq \mathbb{R}$ be a subset of the real numbers.

1. We say $b \in \mathbb{R}$ is a **lower bound** of S if for all $x \in S$, $b \leq x$. If S has a lower bound, we say S is **bounded below**.
2. We say that $l \in \mathbb{R}$ the **greatest lower bound** or **infimum** of S if for all lower bounds b of S, $l \geq b$, and in this case we write
$$l = \inf S.$$
3. If l is the greatest lower bound S and $l \in S$, we say that l is the **least element** of S.

Exercise 4.2. Find the infimum and least element of each set below or say why it doesn't have one.

1. $[0, \infty)$
2. $(-\infty, 0]$
3. $(0, 3]$
4. The set of upper bounds on $[0, 3]$.
5. The set of upper bounds on $[0, 3)$.

Parts 4 and 5 of Exercise 4.2 are meant to demonstrate another way of thinking about the Least Upper Bound Axiom: it tells us that sets of upper bounds *always* have a least element when they are non-empty.

Rather than make an assumption about infima existing, as we did with suprema, we can prove this as a theorem from the Least Upper Bound Axiom using this idea.

Theorem 4.3. *If a nonempty subset $S \subseteq \mathbb{R}$ has a lower bound, then S has a greatest lower bound in $\mathbb{R}$.*

Proof. Suppose $S \subseteq \mathbb{R}$ is nonempty and bounded below. Then let T be the *set of lower bounds* of S. Note that T is nonempty (since S was bounded below) and bounded above (by any element of S). Thus T has a least upper bound $l = \sup T$.

I claim that $l = \inf(S)$. First, we note that l is in fact greater than any other lower bound of S, since l is an upper bound on T. Also, since every element of S is an upper bound on T, we have $l \leq s$ for all $s \in S$ since l is the *least* upper bound of T. □

Exercise 4.4. Try proving Theorem 4.3 another way. Consider the set $U = \{x \ : \ -x \in S\}$, and show that $-\sup U = \inf S$.[1]

One more concept we'll need in order to get started on this section is the idea of taking an infinite union or intersection, which will be important for understanding the Nested Interval Theorem. If you are rusty on unions and intersections, see Section B.2.

Exercise 4.5. For each example below, (i) draw the sequence of intervals on a number line, (ii) decide whether the sequence of intervals satisfies the conditions of Theorem 4.6 below, and (iii) identify the intersection $\bigcap_{i=1}^{\infty} [a_i, b_i]$.

1. $[a_i, b_i] = \left[-\frac{1}{i}, \frac{1}{i}\right]$ for $i \in \mathbb{N}$
2. $[a_i, b_i] = [-i, i]$ for $i \in \mathbb{N}$
3. $[a_i, b_i] = \left[0, 1 + \frac{1}{i}\right]$ for $i \in \mathbb{N}$

[1] We will use this kind of technique frequently to obtain the second two cases of a theorem we want to prove (e.g. one for increasing functions and one for decreasing function, or one for sups and one for infs).

4.2 Nested Interval Theorem

One distinguishing feature of the real numbers is, intuitively, if you keep zooming in on the number line, you are always zooming in *to a number*.

As a thought experiment, consider the following collection of intervals:

$$\begin{aligned} U_1 &= [3, 4] \\ U_2 &= [3.1, 3.2] \\ U_3 &= [3.14, 3.15] \\ U_4 &= [3.141, 3.142] \\ U_5 &= [3.1415, 3.1416] \\ U_6 &= [3.14159, 3.14160] \\ &\vdots \end{aligned}$$

Specifically, U_i is the closed interval between π rounded up and down to i significant figures.

If the universe is $\mathbb{R}$, there is a number that will be in U_i for all i: π. By contrast, in $\mathbb{Q}$, there is no number inside all of these sets—we'd be zooming in onto something that isn't there. The Nested Interval Theorem below generalizes this idea.

Theorem 4.6 (Nested Interval Theorem)**.** *Suppose a_i and b_i are real numbers with $a_i \leq b_i$ for all $i \in \mathbb{N}$ such that if we set $U_i = [a_i, b_i]$, we have $U_{i+1} \subseteq U_i$ for all $i \in \mathbb{N}$. In this case*

$$\bigcap_{i=1}^{\infty} U_i \neq \emptyset.$$

Setup Only. Assuming the hypotheses of the theorem, we note that since $U_{i+1} \subseteq U_i$, we have for all $i \in \mathbb{N}$

$$a_i \leq a_{i+1} \leq b_{i+1} \leq b_i.$$

In fact, from this, one can prove that $a_i \leq b_j$ for any $i, j \in \mathbb{N}$.

… □

Exercise 4.7. Finish the proof of the Nested Interval Theorem.

The big hint here is that Nested Interval Theorem is true for $\mathbb{R}$ but not for $\mathbb{Q}$, so it must use the Least Upper Bound Axiom in some essential way.

Hint

To prove a set is nonempty, you should exhibit an element that lives in the set, *and then prove that element lives in the set.*

4.3 Uncountability of $\mathbb{R}$

Recall that a nonempty set S is called *countable* if there exists a surjective function $\mathbb{N} \to S$. Intuitively, it means that there is some way of listing the elements of S (possibly with repetition) so that every element of S eventually appears on the list.

It is clear that $\mathbb{N}$ is countable, but somewhat less obvious are the facts that $\mathbb{Z}$ and $\mathbb{Q}$ are countable. However, $\mathbb{R}$ is uncountable; it has a strictly larger cardinality than $\mathbb{N}$, $\mathbb{Z}$, and $\mathbb{Q}$. Intuitively, the number of real numbers is a "higher level of infinity" than the number of natural/integer/rational numbers.

There are a few ways to prove that $\mathbb{R}$ is uncountable, but as we should now expect, all of them rely on the Least Upper Bound Axiom since uncountability is a property $\mathbb{R}$ has but $\mathbb{Q}$ does not. We will take the approach of using the Nested Interval Theorem, which in turn relies on the Least Upper Bound Axiom.

Theorem 4.8. *$\mathbb{R}$ is uncountable.*

Proof. By way of contradiction, suppose $\mathbb{R}$ is countable. Then there exists a surjective function $f : \mathbb{N} \to \mathbb{R}$. We will use this function to construct a sequence of nonempty nested closed intervals $\{U_i\}$ and apply the Nested Interval Theorem to arrive at a contradiction. The idea will be to construct U_{i+1} so that it excludes $f(i)$, and that way no value of f will lie in all of the intervals.

Let $a_1 = 0$ and $b_1 = 1$. For each $i \geq 1$, we recursively construct a_{i+1} and b_{i+1} using the following procedure:

- If $f(i) \notin [a_i, b_i]$, we let $a_{i+1} = a_i$ and $b_{i+1} = b_i$ (i.e. the interval remains unchanged).
- If $f(i) \in \left[a_i, \frac{b_i+a_i}{2}\right]$ (that is, $f(i)$ is in the left half of the interval), let $a_{i+1} = \frac{2b_i+a_i}{3}$ and $b_{i+1} = b_i$ (so that $[a_{i+1}, b_{i+1}]$ is the right third of the interval).
- If $f(i) \in \left(\frac{b_i+a_i}{2}, b_i\right]$ (that is, $f(i)$ is in the right half of the interval), let $a_{i+1} = a_i$ and $b_{i+1} = \frac{b_i+2a_i}{3}$ (so that $[a_{i+1}, b_{i+1}]$ is the left third of the interval).

Note that in each of the three cases $f(i) \notin [a_{i+1}, b_{i+1}]$. Letting $U_i = [a_i, b_i]$ for $i \geq 1$, this means that $f(i) \notin U_{i+1}$, and therefore no value $f(n)$ is in *all* of the closed intervals. That is,

for all $n \in \mathbb{N}$,

$$f(n) \notin \bigcap_{i=1}^{\infty} U_i$$

However, by our assumption, every real number is equal to $f(n)$ for some $n \in \mathbb{N}$, so no real numbers are in the intersection, contradicting the Nested Interval Theorem. □

We mention here that the proof above actually shows that $[0, 1]$ is uncountable, and can be adapted to prove the following.

Theorem 4.9. *Any interval $[a, b] \subseteq \mathbb{R}$ with $a < b$ is uncountable.*

Review Questions

1. Give the definitions of lower bound, bounded below, infimum, and least element.
2. What does the Nested Interval Theorem say? Give an example that shows how it distinguishes $\mathbb{R}$ from $\mathbb{Q}$.
3. What does it mean when we say $\mathbb{R}$ is uncountable?
4. How did we know we would need to use the Least Upper Bound Axiom or one of its consequences when proving $\mathbb{R}$ is uncountable?

Part II

Metric Spaces

5 Metric Spaces and Distance Functions

Highlights

- The definitions of "metric space" and "distance function"
- The usual distance functions on $\mathbb{R}^n$, including $\mathbb{R} = \mathbb{R}^1$.
- Properties of absolute values

5.1 Preparation

The limits that you learn about in calculus are often described in terms of a variable getting "closer and closer" or "arbitrarily close" to some quantity. In order to say more precisely what it means to be "close", though, we need to be precise about what we mean by "distance" in the first place.

In this chapter we'll talk about a fundamental construction in the theory of real analysis, which is the concept of a metric space. Basically, a metric space will be any set with a reasonable (and we'll make precise what we mean by "reasonable") notion of distance.

You are likely familiar with the distance formulas in $\mathbb{R}^2$ and $\mathbb{R}^3$, and we will use these two spaces as our mental model for how to think about metric spaces. We'll see that distance can reasonably be defined in other contexts as well, including in the real number system.

For the moment, play around with the applet below to see if you can think about what formula would give us the distance between the real numbers a and b.

Our philosophy will be that we will:

- Gain intuition from $\mathbb{R}^2$ and $\mathbb{R}^3$,
- Write our proofs for any metric space, and
- Apply what we learn to $\mathbb{R}$.

For whatever reason, it's easy for our brains to think about geometry in two and three dimensions than it is in one dimension. And proving theorems about general metric spaces is

sometimes easier than proving the corresponding theorems in $\mathbb{R}$ since there are fewer tools to keep track of[1].

As a side benefit to this approach, the theory of metric spaces is useful more broadly in pure and applied math since we often want to think about how to define the "distance" between objects in a reasonable way. As a teaser of this concept, what is a reasonable way to define the distance between two *functions*?

As we said earlier, we will define a metric space to be any set in which we can reasonably define distance. We begin by making the idea of "reasonableness" precise here.

Definition 5.1. A **metric space** is a nonempty set X equipped with a function $d : X \times X \to \mathbb{R}$ called a **distance function** with the following properties.

1. (Positivity) For any $a, b \in X$, $d(a, b) \geq 0$.
2. (Positive Definiteness) For any $a, b \in X$, $d(a, b) = 0$ if and only if $a = b$.
3. (Symmetry) For any $a, b \in X$, $d(a, b) = d(b, a)$.
4. (Triangle Inequality) For any $a, b, c \in X$, $d(a, b) + d(b, c) \geq d(a, c)$.

Exercise 5.2. Explain why the usual distance formula in $\mathbb{R}^2$,

$$d((x_1, y_1), (x_2, y_2)) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2},$$

satisfies the properties in Definition 5.1. Give a proof for Positivity, Positive-Definiteness, and Symmetry, and provide a picture with an intuitive explanation for the triangle inequality.

While we are used to measuring distances in two and three dimensions, we can define distances in other contexts as well.

Exercise 5.3. Let S be any set. Prove that $d : S \times S \to \mathbb{R}$ defined for any $x, y \in S$ by

$$d(x, y) = \begin{cases} 0 & x = y \\ 1 & x \neq y \end{cases}$$

is a distance function for S (i.e. that it satisfies the four properties). We call this the **trivial** metric for S.

[1]For instance, most metric spaces don't have arithmetic or inequalities, so we won't be distracted by these features of $\mathbb{R}$ in contexts where they aren't relevant.

To talk about a metric space, we really ought to specify both the set X and the distance function d. For example, it's technically possible to equip $\mathbb{R}^2$ with the trivial metric from Exercise 5.3 (although I'm hard-pressed to think of why this would be useful). A somewhat more practical alternative metric on $\mathbb{R}^2$ is the subject of Exercise 5.11, but we'll refer to the distance function from Exercise 5.2 as the "standard" distance function for $\mathbb{R}^2$.

Exercise 5.4. Make a guess about how we will define the "standard" distance function that will turn $\mathbb{R}$ into a metric space.

5.2 Distances in $\mathbb{R}^n$

You are likely familiar with the distance formulas in $\mathbb{R}^2$ and $\mathbb{R}^3$. Generalizing those formulas, the standard distance function in $\mathbb{R}^n$ is given by

$$d((a_1, \dots, a_n), (b_1, \dots, b_n)) = \sqrt{(b_1 - a_1)^2 + \cdots + (b_n - a_n)^2}. \tag{5.1}$$

The proofs that this distance function satisfies Positivity, Positive-Definiteness, and Symmetry are straightforward generalizations of what you did in Exercise 5.2. The proof that the standard distance function satisfies the Triangle Inequality is much more involved, and we defer it to Theorem C.1 in Appendix C.

More important to us at the moment, though, is how to define distances in $\mathbb{R}$. If we apply the formula (5.1) to the case where $n = 1$ we get[2] $d(a, b) = \sqrt{(b - a)^2}$. This ends up simplifying to $|b - a|$, which hopefully matches your guess in Exercise 5.4.

Thus we define the **standard distance function on** $\mathbb{R}$ to be

$$d(a, b) = |b - a|.$$

> From now until the end of your days, whenever you see an absolute value—especially the absolute value of a difference—you should think "distance".

Now that we see absolute values will be our main tool for talking about distances in the real numbers, it will be worthwhile to review some key properties of absolute values.

[2] Here we set $a = (a_1)$, and $b = (b_1)$ to simplify the notation.

5.3 Absolute Values

Since distances in $\mathbb{R}$ are measured using the absolute value function, we'll want to make sure we understand this function and its basic properties well before going much further. The absolute value function is defined for any $x \in \mathbb{R}$ via

$$|x| = \begin{cases} x & x \geq 0 \\ -x & x < 0 \end{cases}$$

Make sure you understand how the second case matches what you know about absolute values before moving on!

Theorem 5.5. *Suppose $a, b \in \mathbb{R}$.*

1. $|a| \geq 0$.
2. $|a| = \max\{a, -a\}$.
3. $|-a| = |a|$.
4. $|a| \geq a$.
5. $|ab| = |a||b|$.
6. $|a + b| \leq |a| + |b|$.
7. $|a - b| \geq \big||a| - |b|\big|$.

Proof. We will prove parts 1–5. Since the definition of absolute value is piecewise, we need to do some case-by-case proofs here.

1. If $a \geq 0$, then $|a| = a \geq 0$. If $a < 0$, then $|a| = -a > 0$.
2. If $a \geq 0$, then $a \geq 0 \geq -a$, so $|a| = a = \max\{a, -a\}$. If $a < 0$, then $-a > 0 > a$, so $|a| = -a = \max\{a, -a\}$.
3. By the previous part, $|a| = \max\{a, -a\} = |-a|$.
4. Follows from the previous part.
5. This can be proven by considering each of the three cases where a, b are both positive, a, b are both negative, and where one of a, b is positive and the other is negative.

□

Exercise 5.6. Prove the remaining parts of Theorem 5.5. As a hint for Part 6, try breaking into two cases: one where $a + b \geq 0$ (so that $|a + b| = a + b$) and one where $a + b < 0$ (so that $|a + b| = (-a) + (-b)$) and then apply Part 4 to both a and b.

Exercise 5.7. Prove that the standard distance function $d(x, y) = |y - x|$ for $\mathbb{R}$ satisfies the Triangle Inequality.

5.4 More Exercises

Exercise 5.8. Let X be a metric space with distance function d and let $x, y \in X$. Suppose that for all $\epsilon > 0$, $d(x, y) \leq \epsilon$. Prove that $x = y$.

> **Tip**
>
> In this section, you should cite Positivity, Positive-Definiteness, and the Triangle Inequalities whenever you use them (don't bother with Symmetry). After this section, you need only cite the Triangle Inequality.

Exercise 5.9. Let X be a metric space with distance function d. Prove that for any $a, b, c \in X$,

$$|d(a, b) - d(b, c)| \leq d(a, c).$$

Exercise 5.10. Prove the "polygon inequality", which says that for any $x_1, \ldots, x_n$ with $n \geq 3$ in a metric space X with distance function d,

$$d(x_1, x_2) + d(x_2, x_3) + \cdots + d(x_{n-1}, x_n) \geq d(x_1, x_n)$$

Hint: this is a good candidate for mathematical induction. It's also a good example of how drawing a picture in two dimensions can help give you intuition about what happens in any metric space.

Exercise 5.11. Here we will consider $\mathbb{R}^2$ with a different distance function. Define

$$d_1((x_1, y_1), (x_2, y_2)) = |x_2 - x_1| + |y_2 - y_1|$$

Prove that d_1 does in fact satisfy the four conditions of being a metric space.

Exercise 5.12. Here we will consider $\mathbb{R}$ with a different purported distance function. Define

$$d_{\text{inv}}(a, b) = \begin{cases} \frac{1}{|a-b|} & a \neq b \\ 0 & a = b \end{cases}$$

Show that d_{inv} is *not* a distance function.

Exercise 5.13. If X is a metric space with distance function d, and $S \subseteq X$ is nonempty, then we can think of S itself as a metric space using the same distance function restricted to S. In this case we say S is a **subspace** of X.

1. If we treat $\mathbb{Q}$ as a subspace of $\mathbb{R}$, prove that the distance between any two points in $\mathbb{Q}$ is a rational number.
2. If we treat $\mathbb{Z}^2$ (the set of points in $\mathbb{R}^2$ both of whose entries are integers), as a subspace of $\mathbb{R}^2$, is the distance between two points in $\mathbb{Z}^2$ always an integer? Explain.

Review Questions

1. State the four conditions a distance function on a metric space must satisfy.
2. What is the standard distance function in $\mathbb{R}^n$? What about in $\mathbb{R}$?
3. Which properties of absolute value from Theorem 5.5 did you already know? Which were new to you?

6 $\mathbb{R}$ as a Metric Space

Highlights

- See how distance interacts with previously established properties of $\mathbb{R}$ (arithmetic, inequalities, and least upper bounds).
- Equivalent conditions to being a least upper bound.
- Use distance to investigate the relationships among the integers, rationals, irrationals, and reals.

6.1 Preparation

In this section, we will focus on thinking about the real number system as a metric space. We start with what will end up being one of the most-cited theorems in the course.

Theorem 6.1. *Let $r > 0$ and $a \in \mathbb{R}$. Then $|a - x| < r$ if and only if $x \in (a - r, a + r)$.*

Exercise 6.2.

1. Draw a picture of what this Theorem 6.1 is saying on the number line.
2. Prove Theorem 6.1.
3. Rewrite $|a - x| < r$ as a statement about distance in $\mathbb{R}$.
4. Rewrite $x \in (a - r, a + r)$ as an inequality.

Theorem 6.1 is foundational because it connects a statement about analysis (distances and closeness) to a statement about geometry (subsets of $\mathbb{R}$). In general, mathematicians love "if and only if" statements because they often let you translate a statement in one context to a statement in another, where you may have a whole new set of tools you can apply. Together with Parts 3 and 4 from Exercise 6.2, you now have four toolbags you can use to analyze the statement: absolute values, intervals, distances, and inequalities.

Indeed, the connections furnished by Theorem 6.1 can be used to help us prove other theorems relating the concepts we've encountered so far.

Theorem 6.3. *Suppose b is an upper bound on a nonempty subset $S \subseteq \mathbb{R}$. Then the following are equivalent.*

1. $b = \sup S$
2. *For all* $\epsilon > 0$ *there exists* $x \in S$ *such that* $x \in (b - \epsilon, b]$.
3. *For all* $\epsilon > 0$ *there exists* $x \in S$ *such that* $|b - x| < \epsilon$.

Proof. Suppose b is an upper bound on S.

$(1 \Rightarrow 2)$: Suppose $b = \sup S$ and let $\epsilon > 0$. Then $b - \epsilon$ is not an upper bound on S because $b - \epsilon < b$ and b is the *least* upper bound on S. That means there exists $x \in S$ with $b - \epsilon < x$. Since b is an upper bound on S, $x \leq b$. Thus we have shown $x \in (b - \epsilon, b]$.

$(2 \Rightarrow 3)$: If $x \in (b - \epsilon, b]$, then $x \in (b - \epsilon, b + \epsilon)$, so $|b - x| < \epsilon$ by Theorem 6.1.

$(3 \Rightarrow 1)$: We will prove this by contraposition. Suppose b is not the least upper bound on S. That means there exists another upper bound u on S such that $u < b$. If we let $\epsilon = b - u$, then for any $x \in S$, $x \leq b - \epsilon = u$, so by Theorem 6.1, $|b - x| \geq \epsilon$. □

> **Note**
>
> When we encounter a new concept, we will often have a TFAE ("the following are equivalent") theorem that follows giving us alternative ways of understanding the new concept. Since these theorems are so useful, I will color-code them with a lightseagreen vertical bar. In fact, the last two theorems are of this kind.

Exercise 6.4. Carefully prove that 5 is the supremum of $[0, 5)$ using Part 2 or 3 of Theorem 6.3.

Exercise 6.5. Write down a "the following are equivalent" theorem about infima analogous to Theorem 6.3 and prove it.

6.2 The max and min of a Finite Set

In Theorem 5.5, I used the notation $\max\{a, -a\}$ to mean whichever of these two numbers is greater. Here we will make that notation precise with some words of caution about when you can use it.

Theorem 6.6. *Suppose* $F \subseteq \mathbb{R}$ *is a nonempty finite set. Then* $\sup F$ *exists and* $\sup F \in F$ *(i.e. that* F *has a greatest element). In this case we write* $\max F$ *for the greatest element of* F.

For example, $\max\{3, 7, 5\} = 7$.

Proof. Let n be the number of elements of F. We will prove that F has a greatest element by induction on n.

As a base case, suppose $n = 1$. Then $F = \{x_1\}$. In this case $x_1 \geq x_1$, so x_1 is an upper bound on F. And any upper bound b on F must have $b \geq x_1$, so x_1 is the least upper bound. And of course $x_1 \in F$.

Now suppose that $k \in \mathbb{N}$ and any set with k elements has a greatest element. Now suppose $F = \{x_1, \ldots, x_k, x_{k+1}\}$ has $k+1$ elements. Define $E = \{x_1, \ldots, x_k\}$ so that E has k elements and therefore a greatest element g. If $x_{k+1} \geq g$, then let $b = x_{k+1}$, and otherwise, let $b = g$. Either way, $b \geq x_{k+1}$ and $b \geq g \geq x_i$ for all $i = 1, \ldots, k$ (so b is an upper bound), and either way $b \in F$. And if u is any upper bound on F, then $u \geq g$ and $u \geq x_{k+1}$, so $u \geq b$. Thus, b is the *least* upper bound of F, and since $b \in F$, b is the greatest element. □

> **Caution**
>
> In this text we only write $\max S$ when S is known to be a *nonempty finite* set. We have seen examples of infinite sets S where S has no greatest element.

Exercise 6.7. Prove that if F is a finite nonempty subset of $\mathbb{R}$, then F has a least element, which we call the min .

Hint: You can try replicating the proof of Theorem 6.6, but it may be easier to *apply* that theorem to F somehow instead!

6.3 Integers, Rationals, and Irrationals in $\mathbb{R}$

By now we may have an intuitive sense that even though the rationals have "holes" where we would expect suprema to be, they are still "everywhere" along the number line. This somewhat bizarre discrepancy is made precise by the next few exercises.

Exercise 6.8. Let $a \in \mathbb{R}$. Prove that there exists a unique $n \in \mathbb{Z}$ such that $n \leq a < n + 1$.

Note that you will need the Well-Ordered Property of $\mathbb{Z}$ in order to prove this (see Appendix D).

Exercise 6.8 now allows us to use the "floor function"; for any $x \in \mathbb{R}$ we can define $\lfloor x \rfloor$ to be the greatest integer that is less than or equal to x; similarly, $\lceil x \rceil$, the "ceiling", is the least integer that is greater than or equal to x. In practice, students tend to overuse the floor function in real analysis proofs—most of the time you just need *an* integer that is greater x, which is furnished by the Archimedean Property (Theorem 3.8).

Exercise 6.9. Let $b \in \mathbb{R}$ and $q \in \mathbb{N}$. Prove that there exists a unique $p \in \mathbb{Z}$ such that

$$\frac{p}{q} \leq b < \frac{p+1}{q}$$

> **Hint**
>
> Don't start from scratch. How can you *apply* the previous exercise to help you prove the result of this one? Here we need to prove something about *all* $b \in \mathbb{R}$ and *all* $q \in \mathbb{N}$, so we must start with something like "Suppose $b \in \mathbb{R}$ and $q \in \mathbb{N}$". To what value of a should we apply the result of Exercise 6.8?

Exercise 6.10. Let $b \in \mathbb{R}$ and $\epsilon > 0$. Prove that there exists $r \in \mathbb{Q}$ such that $d(b, r) < \epsilon$.

> **Hint**
>
> Once again, don't start from scratch. First note that the inequality you are trying to prove appears in Theorem 6.1 although the variable names have changed. How can you apply Exercise 6.9 to get what you want?
> If you get stuck, draw this a picture on the number line.
> If you are still stuck, use the following concrete example to help you wrap your head around the problem. If $b = \pi$ and $\epsilon = 0.1$, how would you choose the rational number r? Can you frame your technique in terms of the notation Exercise 6.9? Given ϵ, to what value of q would you apply the result?

Exercise 6.11. Let $c, d \in \mathbb{R}$ with $c < d$. Prove that there exists a rational number x with $c < x < d$.

> **Hint**
>
> Once again, don't start from scratch. In both this exercise and Exercise 6.10, you need to show that a rational number exists in a particular interval but the intervals are described differently. Given c, d, can you figure out what b, ϵ to use so you can apply Exercise 6.10?

Exercise 6.12. Let $c, d \in \mathbb{R}$ with $c < d$. Prove that there exists an irrational number x with $c < x < d$.

> **Hint**
>
> Think about using Theorem 4.9 and a cardinality argument here!

Review Questions

1. A good study strategy for this course is to keep a catalog of all the major concepts in the course, and then quiz yourself by picking two at random and asking yourself to think of how they are connected.[1] Give this a try for the major concepts we've encountered so far

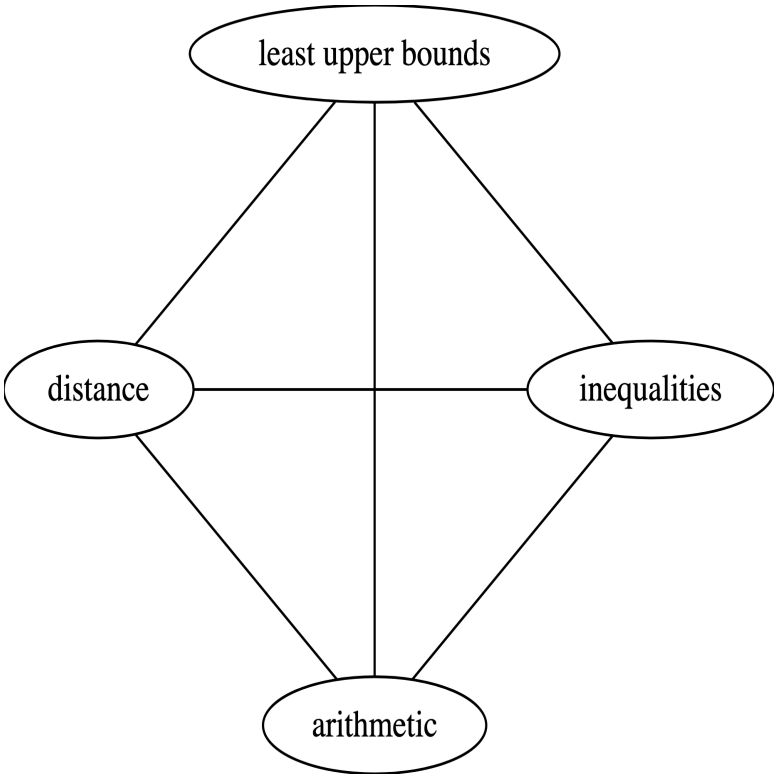


Figure 6.1

2. What do we mean when we write $\max S$? When are we "allowed" to use this notation?
3. In what sense are the rationals "all over" the real number line?

[1] I heard one study skills expert suggest writing the concepts on index cards, shuffling, and drawing the cards two at a time.

7 Open Sets

Highlights

- Definition of an open ball and open set
- Important equivalent condition to being an open set
- Examples of proving and applying openness

7.1 Preparation

An important tool for describing "closeness" precisely in calculus is the open ball.

Definition 7.1. Suppose X is a metric space with distance function d. Let $a \in X$ and $r > 0$. Then the **open ball** centered at[1] a of radius r is defined to be

$$\mathrm{B}_r(a) = \{x \in X \ : \ d(a,x) < r\}.$$

Other textbooks sometimes call $\mathrm{B}_r(a)$ an "r-neighborhood" of a.

Exercise 7.2. Describe and draw pictures of open balls in $\mathbb{R}^3$, $\mathbb{R}^2$, and $\mathbb{R}$.

Exercise 7.3. Answer the following questions about open balls in $\mathbb{R}$.

1. Write $\mathrm{B}_r(c)$ in interval notation.
2. If $a, b \in \mathbb{R}$ with $a < b$, is the interval (a, b) an open ball? If so, what is its center and radius?
3. Is $(0, \infty)$ an open ball? If so, what is its center and radius?

The following exercise is meant to prepare you for the definition of an open set.

Exercise 7.4. Say whether each statement below is true or false and give a brief justification.

1. "If $S = (1, \infty)$, for each point $a \in S$ there is an open ball centered at a that lies entirely inside of S."
2. "If $S = [1, \infty)$, for each point $a \in S$ there is an open ball centered at a that lies entirely inside of S."

[1] Sometimes we also say "around a".

7.2 Open Balls

When we apply the definition of an open ball to $\mathbb{R}$, we realize that it's an object we've dealt with before; in particular Theorem 6.1 was really a statement about open balls on the real number line.

Putting Theorem 6.1 together with the language of open balls, we now have many equivalent ways of saying the same thing:

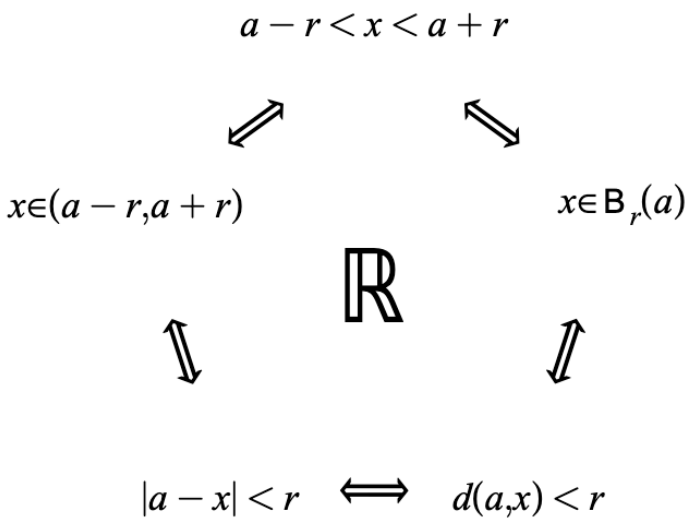


Figure 7.1: Equivalent formulations for being an open ball in the metric space $\mathbb{R}$.

The equivalent conditions of Figure 7.1 in $\mathbb{R}$ will show up every day for the rest of the course. The more seamlessly you can translate among these different ways of saying the same thing, the easier the course will be.

Becoming fluent in Figure 7.1 is the cheat code for reading analysis proofs.

While using the word "ball" makes intuitive sense in $\mathbb{R}^3$ (and sort of in $\mathbb{R}^2$), it may feel a little weird in $\mathbb{R}$; we'll just need to get used to that. And it gets even stranger when we think about other metric spaces.

Exercise 7.5. Let X be a set with the trivial metric (defined in Exercise 5.3). Let $a \in X$, and describe the following.

1. $B_2(a)$
2. $B_1(a)$
3. $B_{1/2}(a)$

Exercise 7.6. Consider $\mathbb{R}^2$ with the "taxicab metric" d_1 defined in Exercise 5.11. Sketch the open ball of radius 2 centered at the origin.

Exercise 7.7. Consider $S = \{(x, y) \ : \ y \geq 0\}$ as a subspace of $\mathbb{R}^2$. Sketch $\mathrm{B}_2(0, 1)$ in the metric space S.

Exercise 7.8. Suppose X is a metric space with distance function d. Let a, b be distinct points in X. Prove that there exists $\epsilon > 0$ such that $\mathrm{B}_\epsilon(a)$ and $\mathrm{B}_\epsilon(b)$ are disjoint.

7.3 Open Sets

Now we transition from thinking about open balls to the more general concept of an "open set".

Definition 7.9. Let X be a metric space with distance function d. A subset $S \subseteq X$ is called *open* if every point $a \in S$ has an open ball around it that lies entirely within S. That is, for all $a \in S$, there exists $r > 0$ such that $\mathrm{B}_r(a) \subseteq S$.

Intuitively, an open set S cannot include any of its boundary points because an open ball around a boundary point will contain points outside of S.

Exercise 7.10. Let X be a metric space with distance function d. Prove that X and $\emptyset$ are open subsets of X.

Exercise 7.11. Prove that open balls are open.

Exercise 7.12. Suppose X is a metric space and let $a \in X$. Prove that $\{a\}^\mathsf{C}$ is open.

Exercise 7.13. Prove that the interval $(-\infty, 3)$ is open in $\mathbb{R}$.

Review Questions

1. What is the definition of an open ball in a metric space?
2. What is the definition of an open set? Can you draw a picture representing this definition?
3. What are the open balls in $\mathbb{R}$?

8 Properties of Open Sets

Highlights

- Characterization of open sets as unions of open balls
- Unions and intersections of open sets

8.1 Preparation

We'll start with a few warm-up exercises using the definition of an open set from Chapter 7.

Exercise 8.1.

1. Write down what it means for a subset of a metric space to *not* be open.
2. Write down a careful proof that $(-\infty, 2]$ is not an open subset of $\mathbb{R}$.

> **Caution**
>
> We are *not* going to call non-open sets "closed". The term "closed" will have a different definition that we'll get to later.

Exercise 8.2. Let U be an open subset of $\mathbb{R}$. Prove that U has no greatest element.

The next theorem gives us another useful characterization of open sets—that each is a union of a collection of open balls.

Theorem 8.3. *Let X be a metric space and $U \subseteq X$. The following are equivalent.*

1. *U is open.*
2. *U is a union of open balls.*

Theorem 8.3 is another "the following are equivalent" theorem. We love these because anytime there is a proof involving an open set, we get to pick which condition to use. Before getting into the proof of this theorem, we'll see how it can be used.

Exercise 8.4. Redo Exercise 7.13 using Condition 2 of Theorem 8.3 by showing that the interval $(-\infty, 3)$ is a union of open balls.

8.2 Open Sets as Unions of Open Balls

Now we get to the proof.

Exercise 8.5. Prove Theorem 8.3.

It is worth mentioning that some open sets are easier to witness as unions of open balls than others. Compare the two examples below.

Exercise 8.6. Prove that each set below is open. Think about which condition of Theorem 8.3 will be easiest to use in each case.

1. The complement of $\mathbb{Z}$ in $\mathbb{R}$.
2. The upper half plane $H = \{(x, y) \in \mathbb{R}^2 \ : \ y > 0\}$ as a subset of $\mathbb{R}^2$.

8.3 Unions and Intersections of Open Sets

We can construct new open sets out of old ones using unions and intersections via the following theorem.

Theorem 8.7.

1. *The union of any collection of open sets is open.*
2. *The intersection of any finite collection of open sets is open.*

Exercise 8.8. Prove Part 1 of Theorem 8.7.

Hint: Will it be easier to work with the *definition* of openness, or Condition 2 of Theorem 8.3 for this proof?

Setup of proof of Part 2 of Theorem 8.7. Suppose $U_1, \ldots, U_n$ are open and let $U = \bigcap_{i=1}^{n} U_i$. We will prove U is open using the definition of openness, so we start by supposing that $x \in U$. This means $x \in U_i$ for all $i = 1, \ldots, n$. Therefore, applying the definition of openness to each U_i, there exists r_i such that $\mathrm{B}_{r_i}(x) \subseteq U_i$. □

Exercise 8.9. Finish the proof of Part 2 of Theorem 8.7.

Exercise 8.10.

1. At what point in the proof of Part 2 of Theorem 8.7 did you use the fact that we were only dealing with *finitely many* open sets?
2. Give an example of a(n infinite) collection of open sets whose intersection is *not* open. (Remember, $\emptyset$ is open!)

Review Questions

1. Say what it means for a set to be "not open".
2. What are the equivalent characterizations of open sets?
3. How do open sets behave under unions and intersections?

9 Boundedness

Highlights

- Definition of a bounded subset of a metric space
- Equivalent conditions to boundedness
- Boundedness in $\mathbb{R}$ is equivalent to "bounded above and below"

9.1 Preparation

Theorem 9.1. *Let X be a metric space and let $S \subseteq X$. Then the following are equivalent.*

1. *For any $a \in X$, there exists $r > 0$ such that $S \subseteq \mathrm{B}_r(a)$.*
2. *There exists $a \in X$ and $r > 0$ such that $S \subseteq \mathrm{B}_r(a)$.*

*In this case, we say S is **bounded**.*

You will prove Theorem 9.1 in Exercise 9.4 below.

> **Note**
>
> Theorem 9.1 is both a theorem and a definition. Once we prove the conditions are equivalent, we can refer to any one of them as "the" definition of bounded.

Exercise 9.2.

1. If a problem asks you to *prove* that a set is bounded, which of the two conditions from Theorem 9.1 would you choose to work with, and why?
2. If a problem tells you to assume that a set is bounded and you need to *apply* boundedness in order to prove something else, which of the two conditions from Theorem 9.1 would you choose to work with, and why?

Exercise 9.3. Negate each of the two equivalent characterizations of boundedness.

A subset of a metric space that is not bounded is called **unbounded**.

9.2 Properties of Boundedness

We start with proving that the two definitions of boundedness are in fact equivalent.

Exercise 9.4. Prove Theorem 9.1.

> Hint
>
> 1. One of the two implications should be very quick to prove!
> 2. Note that a might lie outside of S for either of the two conditions.

Exercise 9.5. Let $S = \{(n, n) \in \mathbb{R}^2 \ : \ n \in \mathbb{N}\}$. Draw a picture of S and give a careful proof that S is unbounded.

Exercise 9.6. Suppose S_1 and S_2 are bounded subsets of a metric space X. Prove that $S_1 \cup S_2$ is bounded.

9.3 Boundedness in $\mathbb{R}$

As ever, what we really care about is how this concept applies to $\mathbb{R}$ and how the concepts defined thus far interact. We've already defined "bounded above" and "bounded below". The next theorem will connect these definitions to our new more general definition of "bounded" for metric spaces.

Theorem 9.7. *Suppose $S \subseteq \mathbb{R}$. Then the following are equivalent.*

1. *S is bounded.*
2. *S has an upper bound and a lower bound.*
3. *There exists $K > 0$ such that for all $x \in S$, $|x| < K$.*

Exercise 9.8. Prove Theorem 9.7.

Exercise 9.9. Prove that the complement of a bounded subset of $\mathbb{R}$ is unbounded.

Review Questions

1. What does it mean for a subset of a metric space to be bounded? Unbounded?
2. With Theorem 9.1 and Theorem 9.7, we have four equivalent ways of describing boundedness in $\mathbb{R}$; what are they?
3. Building on Review Question 1 from Chapter 6, think about what theorems or facts relate the major concepts we've encountered thus far.

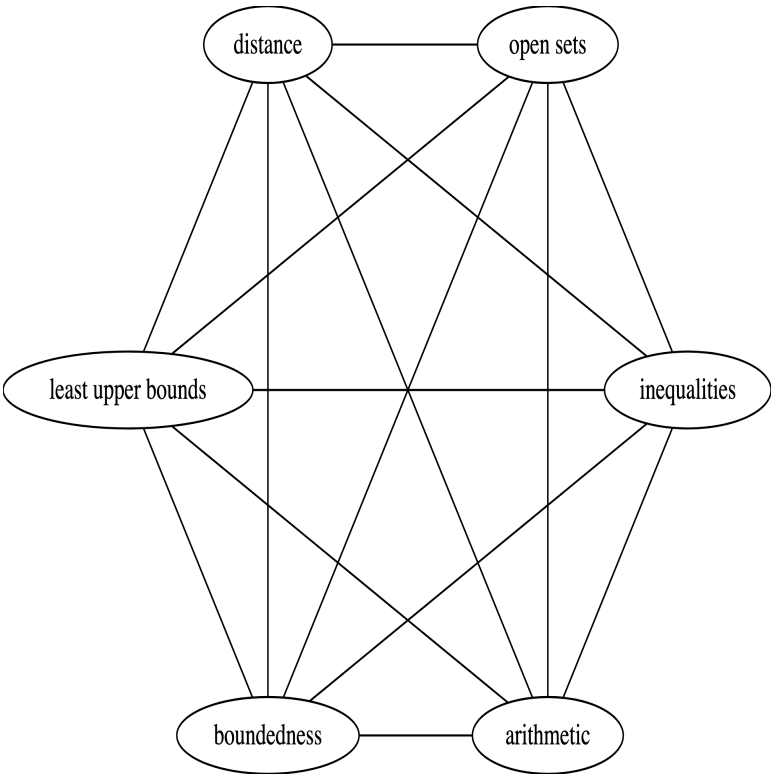

distance
open sets
least upper bounds
inequalities
boundedness
arithmetic

Figure 9.1

Part III

Assorted Topics (Provisional)

10 Series Basics

Highlights

- Definitions of series, partial sums, and series convergence
- Examples of series converging and diverging
- Special properties of series with all non-negative terms.

10.1 Preparation

In this section, we build on what we learned about sequences to talk about one of their most important use cases: series. Loosely speaking, we think of a series as an "infinite sum", and one of the primary questions we seek to address is when a series converges to a finite number and when it doesn't. Consider for example, the four infinite sums below:

$$\begin{aligned} &1+2+3+4+\cdots \\ &1+(-1)+1+(-1)+\cdots \\ &1+\tfrac{1}{2}+\tfrac{1}{3}+\tfrac{1}{4}+\cdots \\ &1+\tfrac{1}{2}+\tfrac{1}{4}+\tfrac{1}{8}+\tfrac{1}{16}+\cdots \end{aligned} \tag{10.1}$$

Some of these might strike you as obviously undefined or obviously convergent, but others may not. In what follows below, we'll make a precise definition of what we mean by convergent so that we can prove theorems and talk coherently about examples.[1]

Suppose (a_n) is a sequence in $\mathbb{R}$. Then the **series** with terms (a_n) is the formal expression

$$\sum_{i=1}^{\infty} a_i = a_1 + a_2 + \cdots$$

[1]There are actually several theories of convergent series which are useful in different contexts. We'll present the most standard version here.

By "formal expression" I mean that we are not considering this expression to represent a number; it's just notation meant to express the *idea* of an infinite sum. Sometimes we abbreviate the notation as $\sum a_i$.

Definition 10.1 (Partial Sums)**.** The **sequence of partial sums** of the series $\sum_{i=1}^{\infty} a_i$ is the sequence (s_n) defined by

$$s_n = \sum_{i=1}^{n} a_i = a_1 + \cdots + a_n.$$

This means that

$$\begin{aligned} s_1 &= a_1 \\ s_2 &= a_1 + a_2 \\ s_3 &= a_1 + a_2 + a_3 \\ &\vdots \end{aligned}$$

We now define convergence of series in terms of the partial sums.

Definition 10.2 (Series Convergence)**.** The series $\sum_{i=1}^{\infty} a_i$ is said to **converge** or **diverge** when its sequence of partial sums (s_n) does. If $s_n \to S$, we call S the **sum** of the series.

In the case that $\sum_{i=1}^{\infty} a_i$ has sum S, we sometimes abuse notation slightly and use the expression $\sum_{i=1}^{\infty} a_i$ to represent the number S.

As an example of the vocabulary in action, the series $\sum_{i=1}^{\infty} \frac{2}{3^i}$

- has *terms* given by the sequence $\left(\frac{2}{3^i}\right)$ or $\frac{2}{3}, \frac{2}{9}, \frac{2}{27}, \frac{2}{81}, \cdots$
- has *partial sums* $\frac{2}{3}, \frac{2}{3} + \frac{2}{9}, \frac{2}{3} + \frac{2}{9} + \frac{2}{27}, \ldots$. These simplify to $\frac{2}{3}, \frac{8}{9}, \frac{26}{27}, \ldots$ or $\left(\frac{3^n - 1}{3^n}\right)$.
- *converges* with a *sum* of 1.

Exercise 10.3. Write each of the four infinite sums in Equation 10.1 using series notation. Then write out the first 6 partial sums for each. Are there any you can tell right away converge or diverge?

10.2 Working with Partial Sums

We start with an important class of series called *geometric series*.

Example 10.4 (Geometric Series)**.** For $a \neq 0$ and any $r \in \mathbb{R}$, a series of the form

$$\sum_{i=0}^{\infty} ar^i$$

is called a **geometric series**. This series converges to $\frac{a}{1-r}$ when $|r| < 1$ and diverges otherwise.

Proof. To determine whether the series converges, we analyze its partial sums in cases.

If $|r| \neq 1$, we can use some algebraic tricks to simplify the expression for the partial sum s_n :

$$\begin{aligned} s_n &= a + ar + ar^2 + \cdots + ar^n \\ &= a(1 + r + r^2 + \cdots + r^n) \\ &= \frac{a(1 - r^{n+1})}{1 - r} \end{aligned}$$

If $|r| < 1$, then $r^{n+1} \to 0$, so using what we know about sequences and arithmetic, we can say $s_n \to \frac{a}{1-r}$.

If $|r| > 1$, then (r^{n+1}) diverges. So if, by way of contradiction, we assume the partial sums (s_n) did converge to some value S, then we could say that the sequence $\left(1 - s_n \frac{1-r}{a}\right)$ converges to $1 - S\frac{1-r}{a}$. However $1 - s_n \frac{1-r}{a} = r^{n+1}$, which we already said diverges.[2]

We leave the cases where $r = \pm 1$ to the reader in Exercise 10.5 below. □

Exercise 10.5. Finish the proof of Example 10.4 by proving that the geometric series diverges when $r = 1$ and when $r = -1$. (Hint: what are the partial sums in each of these cases?)

> **Note**
>
> In the section on sequences, we insisted that the first term in our sequence corresponded to an index of 1. For series, it is often convenient to start with an index 0 (as in Example 10.4) or something else. And now that we are more comfortable with the concept of a sequence, there is less risk of confusion.

[2] I am bending over backwards here to make precise the intuition that "Since (r^{n+1}) diverges, so must a sequence involving it". Unfortunately, that reasoning doesn't always work—think about $(\sqrt{n+1} - \sqrt{n})$ which converges despite being the sum of two divergent sequences—so a proof of the style shown here is appropriate.

Exercise 10.6 (A "Telescoping" Series)**.** Show that the series $\sum_{i=1}^{\infty} \frac{1}{i(i+1)}$ converges. It will be helfpul to note that

$$\frac{1}{i(i+1)} = \frac{1}{i} - \frac{1}{i+1}$$

and to write out the first several partial sums.

Both Example 10.4 and Exercise 10.6 are somewhat unusual in that in both cases we can write down an explicit formula for the partial sum sequence. More often an explicit formula is unavailable and we need to rely on other means for determining and proving the con/divergence of a series.

The so-called "convergence tests" for series that we learn in calculus give us tools for determining the con/divergence of a series by analyzing their *terms* rather than their *partial sums*.

Theorem 10.7 (Divergence Test)**.** *If a series converges, then its terms converge to* 0.

Exercise 10.8.

1. State the contrapositive of the Divergence Test.
2. What (if anything) does the Divergence Test say about the con/divergence of the series $\sum_{i=1}^{\infty} \frac{i}{i+1}$?
3. What (if anything) does the Divergence Test say about the con/divergence of the series $\sum_{i=1}^{\infty} \frac{1}{i^2}$?
4. Prove the Divergence Test.

> Hint for part 4 of Exercise 10.8
>
> Note that $a_{n+1} = s_{n+1} - s_n$.

Theorem 10.9. *Suppose* $\sum a_n = S$ *and* $\sum b_n = T$ *are convergent series and that* $k \in \mathbb{R}$*. Then*

1. $\sum(a_n + b_n) = S + T$
2. $\sum(ka_n) = kS$
3. *If* $a_n \leq b_n$ *for all* n*, then* $S \leq T$*.*

Note that part of what parts 1 and 2 in Theorem 10.9 are claiming is that these two series do in fact converge.

> **Note**
>
> When an equation or proposition does not depend on the starting index of a series, we sometimes just write $\sum a_n$ as shorthand for $\sum_{n=k}^{\infty} a_n$ for some k.

Exercise 10.10. Prove Theorem 10.9.

> **Hint**
>
> Leverage what we already know about sequence convergence to do these proofs!

10.3 Series With Non-Negative Terms

Series with all non-negative terms have some special properties that we can quickly prove using what we know about sequences.

Exercise 10.11. Suppose $\sum_{i=1}^{\infty} a_i$ is a series all of whose terms are non-negative. Prove the following.

1. The sequence of partial sums is increasing.
2. The series converges if and only if the sequence of partial sums is bounded.
3. If the series diverges, then the sequence of partial sums limits to $+\infty$.

The upshot of Exercise 10.11 is that we can determine the con/divergence of a series just by knowing whether its sequence of partial sums is bounded or not.

Example 10.12. The series $\sum_{i=1}^{\infty} \frac{1}{i^2}$ converges.

Proof. Let (s_n) be the sequence of partial sums for $\sum_{i=1}^{\infty} \frac{1}{i^2}$.

In Exercise 10.6, we worked with the series $\sum_{i=1}^{\infty} \frac{1}{i(i+1)}$. Let (t_n) be its sequence of partial sums.

As we saw, the series converges to some number T. Therefore by Theorem 10.9, the series $\sum_{i=1}^{\infty} \frac{2}{i(i+1)}$ converges to $2T$; that is $2t_n \to 2T$. Comparing the terms of the two series, we see that for all i,

$$\frac{1}{i^2} \leq \frac{2}{i(i+1)}.$$

Adding up this expression for $i = 1, \ldots, n$, we have that $s_n \leq 2t_n$ for all n. Since $\sum_{i=1}^{\infty} \frac{2}{i(i+1)}$ converges, $(2t_n)$ is bounded above by Exercise 10.11. Therefore (s_n) is also bounded above and thus converges, again by Exercise 10.11. □

Note that the above proof does not tell us what the series $\sum_{i=1}^{\infty} \frac{1}{i^2}$ converges *to*. This is often the case for such proofs.

The following theorem generalizes the logic of the proof of Example 10.12.

Theorem 10.13 (Comparison Test)**.** *Suppose $\sum a_n$ and $\sum b_n$ are series with all non-negative terms such that $a_n \leq b_n$ for all n.*

1. *If $\sum b_n$ converges, then $\sum a_n$ converges.*
2. *If $\sum a_n$ diverges, then $\sum b_n$ diverges.*

Example 10.14 (Harmonic Series)**.** The *harmonic series* $\sum_{i=1}^{\infty} \frac{1}{i}$ diverges.

Proof. Call the terms of this series $c_i = \frac{1}{i}$ and the partial sums s_n. Note that $c_i \to 0$, so the Divergence Test cannot be used to prove divergence.

Consider another series $\sum_{i=1}^{\infty} b_i$ where the terms b_i are defined by

$$\begin{aligned} b_1 &= 1 \\ b_2 &= \tfrac{1}{2} \\ b_3 = b_4 &= \tfrac{1}{4} \\ b_5 = \cdots = b_8 &= \tfrac{1}{8} \\ &\vdots \\ b_{2^{i-1}+1} = \cdots = b_{2^i} &= \tfrac{1}{2^i} \\ &\vdots \end{aligned}$$

Calling the partial sums t_n, we can consider the subsequence t_{2^n} :

$$\begin{array}{rcll} t_{2^1} & = & 1+\frac{1}{2} & = \frac{3}{2} \\ t_{2^2} & = & 1+\frac{1}{2}+2\cdot\frac{1}{4} & = \frac{4}{2} \\ t_{2^3} & = & 1+\frac{1}{2}+2\cdot\frac{1}{4}+4\cdot\frac{1}{8} & = \frac{5}{2} \\ & \vdots & & \end{array}$$

Thus $t_{2^n} = \frac{n+2}{2}$, so (t_n) has a divergent subsequence and hence $\sum_{i=1}^{\infty} b_i$ diverges. Therefore since $c_i \geq b_i$ for all i (and both series have all non-negative terms) $\sum_{i=1}^{\infty} \frac{1}{i}$ diverges as well by the Comparison Test. □

Exercise 10.15. Prove Theorem 10.13.

Review Questions

1. How is series convergence defined?
2. In basically all of the series convergence proofs from this chapter, the first step is rewriting the statement as a statement about the convergence of what *sequence*?
3. What are some special properties of series with all non-negative terms?

11 The Min-Max Theorem

Highlights

- Special properties of how continuous functions interact with closed and bounded subsets of $\mathbb{R}$.

11.1 Preparation

The goal of this section is to prove the Min-Max Theorem, which says that for any nonempty, closed, and bounded subset $C \subseteq \mathbb{R}$ and any continuous function $f : C \to \mathbb{R}$, f achieves a global max and a global min on C. That is, there are points $a, b \in C$, called the *global min* and *global max* respectively, such that for all $x \in C$,

$$f(a) \leq f(x) \leq f(b).$$

Exercise 11.1. Show by example that both closedness and boundedness are necessary conditions for guaranteeing a global max and global min. That is:

1. Find a closed subset C of $\mathbb{R}$ (which is not bounded) and a continuous function $f : C \to \mathbb{R}$ so that f has no global max and/or no global min on C.
2. Find a bounded set C of $\mathbb{R}$ (which is not closed) and a continuous function $f : C \to \mathbb{R}$ so that f has no global max and/or no global min on C.

As we'll see, closed and bounded subsets of $\mathbb{R}$ play an important role, so here are a couple more theorems about such subsets.

Theorem 11.2. *Let C be a closed and bounded subset of $\mathbb{R}$, and let (a_n) be a sequence in C. Then (a_n) has a subsequence that converges to a point in C.*

Exercise 11.3. Prove Theorem 11.2. Note that you not only need to show that (a_n) has a convergent subsequence but also that the limit of that convergent subsequence *is in* C.

11.2 Closed and Bounded Subsets of $\mathbb{R}$

The main theoretical result of this section will be Theorem 11.4, which tells us that if $C \subseteq \mathbb{R}$ is closed and bounded and $f : C \to \mathbb{R}$ is continuous, then the image $f(C)$ is also closed and bounded. As you saw in Exercise 11.3, Bolzano-Weierstrass will serve as the foundation for much of what we do, so we should expect results that hold in $\mathbb{R}$ but not in $\mathbb{Q}$.

Theorem 11.4. *Let C be a closed and bounded subset of $\mathbb{R}$, and suppose $f : C \to \mathbb{R}$ is continuous. Then the image $f(C)$ is also closed and bounded.*

Before we see the proof, it's worth noting that closed and bounded work together here. It's not true that the continuous image of any closed set is closed, nor that the continuous image of any bounded set is bounded, as the exercise below shows.

Exercise 11.5. The following examples show that Theorem 11.4 can fail if either condition on C (closed or bounded) is violated.

1. Let $C = [1, \infty)$ and $f : C \to \mathbb{R}$ be defined by $f(x) = \arctan(x)$. What is $f(C)$? Explain how this example shows that the image of a closed subset of $\mathbb{R}$ need not be closed.
2. Let $C = (0, 1]$ and $f : C \to \mathbb{R}$ be defined by $f(x) = \frac{1}{x}$. What is $f(C)$? Explain how this example shows that the image of a bounded subset of $\mathbb{R}$ need not be bounded.

Proof. Proof of Theorem 11.4.

Suppose C is closed and bounded and that $f : C \to \mathbb{R}$ is continuous. First we will prove that $f(C)$ is closed by showing that the limit of every convergent sequence in $f(C)$ is in $f(C)$. So suppose (y_n) is a convergent sequence in $f(C)$. Let's call its limit b. Since $y_n \in f(C)$, there exists $x_n \in C$ such that $f(x_n) = y_n$ for every $n \in \mathbb{N}$.[1] Then by Theorem 11.2, (x_n) has a subsequence (x_{i_n}) that converges to a point in C; we'll call that point a. Since f is continuous and $x_{i_n} \to a$, we have $f(x_{i_n}) \to f(a)$; that is, $y_{i_n} \to f(a)$. But since $y_n \to b$, we know $y_{i_n} \to b$, so by the uniqueness of sequence limits, we have $b = f(a)$. Since $a \in C$, this proves $b \in f(C)$ as desired.

Now by way of contradiction, suppose $f(C)$ is not bounded. Then there exists a sequence (y_n) in $f(C)$ with no convergent subsequence. As before, since $y_n \in f(C)$, there exists $x_n \in C$ such that $f(x_n) = y_n$ for all $n \in \mathbb{N}$. Again by Theorem 11.2, (x_n) has a subsequence (x_{i_n}) that converges to some point a in C. Since f is continuous and $x_{i_n} \to a$, we have $y_{i_n} \to f(a)$, contradicting the assumption that (y_n) has no convergent subsequence. □

[1] We aren't assuming f is one-to-one, so x_n need not be unique.

Exercise 11.6. This exercise is meant to help you see what's going on in the first part of the proof of Theorem 11.4.

Let $C = [-2, 2]$ and define $f : C \to \mathbb{R}$ by $f(x) = x^2$. Let (y_n) be the sequence $3, 3.9, 3.99, 3.999 \ldots$.

1. What is $f(C)$ in this case? Is (y_n) in $f(C)$?
2. Find a sequence (x_n) in C so that (a) $f(x_n) = y_n$, and (b) (x_n) does not converge. (Hint: you'll need to take advantage of the fact that f is not injective.)
3. Does the sequence (x_n) you chose have a convergent subsequence? If so, what does it converge to?

Lemma 11.7. *Let $C \subseteq \mathbb{R}$ be non-empty, closed, and bounded. Then $\sup C \in C$ and $\inf C \in C$.*

The proof is part of Exercise 11.9 below.

Theorem 11.8 (Min-Max Theorem)**.** *Let C be a non-empty, closed, and bounded subset of $\mathbb{R}$ and suppose $f : C \to \mathbb{R}$ is continuous. Then there exist points $a, b \in C$, which we call the* global min *and* global max *respectively, such that for all $x \in C$*

$$f(a) \leq f(x) \leq f(b).$$

Exercise 11.9.

1. Prove Lemma 11.7.
2. Use Lemma 11.7 to help you prove Theorem 11.8.

We conclude with another consequence of Theorem 11.2 that we'll need for later on in the course.

Theorem 11.10. *Let $C \subseteq \mathbb{R}$ be non-empty, closed, and bounded and let $f : C \to \mathbb{R}$ be a function. Then f is uniformly continuous if and only if f is continuous.*

Proof. We already know that uniform continuity implies continuity, so we just need to prove the backwards implication.

By way of contradiction, suppose $f : C \to \mathbb{R}$ is continuous but not uniformly continuous. Then there exists $\epsilon > 0$ and sequences (x_n) and (y_n) in C such that $|x_n - y_n| \to 0$ and yet for all $n \in \mathbb{N}$,

$$|f(x_n) - f(y_n)| \geq \epsilon. \tag{11.1}$$

Then by Theorem 11.2, the sequence (x_n) has a subsequence (x_{i_n}) that converges to some point $a \in C$. Now consider the corresponding subsequence (y_{i_n}) of (y_n). Since $|x_n - y_n| \to 0$, the subsequence $|x_{i_n} - y_{i_n}| \to 0$ and therefore $x_{i_n} - y_{i_n} \to 0$. Then $y_{i_n} = x_{i_n} - (x_{i_n} - y_{i_n})$, so $y_{i_n} \to a - 0 = a$.

So since f is continuous and $x_{i_n} \to a$, we know $f(x_{i_n}) \to f(a)$. Similarly, since $y_{i_n} \to a$, we know $f(y_{i_n}) \to f(a)$. Therefore $f(x_{i_n}) - f(y_{i_n}) \to 0$, so there exists $M \in \mathbb{N}$ such that for all $m > M$, $|f(x_{i_m}) - f(y_{i_m})| < \epsilon$, contradicting Equation 11.1. □

Exercise 11.11. We proved earlier that if $C \subset \mathbb{R}$ is nonempty, closed, and bounded, and $f : C \to \mathbb{R}$ is continuous and injective, then $f^{-1} : f(C) \to C$ is continuous. Give a new proof that uses Theorem 11.4 (and what we know about inverse images of open sets under continuous maps).

11.3 Review Questions

1. Give examples to show that the conditions of *boundedness* and *closedness* "work together" in Theorem 11.4 and Theorem 11.8 by showing that neither property on its own is sufficient to imply the conclusions of the theorems. Include pictures.
2. Note that many of the theorems above are marked with maroon bars, indicating that they rely (one way or another) on the Least Upper Bound Axiom. Give examples to show that these would be false if our universe were $\mathbb{Q}$ instead of $\mathbb{R}$.
3. Define a subset $U = \{1, \frac{1}{2}, \frac{1}{3}, ...\} \cup \{0\}$. Suppose $f : U \to \mathbb{R}$ is continuous. Prove that there exists $c \in U$ such that $f(c) \geq f(x)$ for all $x \in U$.
4. Since we are seeing the previous problem in this section, it's clear what tools to use. But out of context, it would be harder. Write another example of a problem that tests this content that would be tricky if you saw it out of context.

Part IV

Integration

12 Naive integration

Highlights

- A review of the naive Calc I definition of the integral
- Why basing a theory on the naive definition is insufficient.

12.1 Preparation

Most calculus students strongly associate integrals with taking antiderivatives, and for good reason. Typically, the most efficient way to calculate the exact value of a definite integral is to employ the Fundamental Theorem of Calculus, which tells us how to use the antiderivative of the integrand. However, in order to understand how definite integrals can be applied in both abstract mathematics and in the real world, we should not lose sight of the definition of the definite integral in terms of Riemann sums.

We will start with a review of how the definite integral is presented in a typical Calc I course, what we'll call the "naive integral". However, as we shall see, this definition fails to produce a self-contained theory of integration that behaves nicely.

As a result, it will be impossible to prove the Fundamental Theorem of Calculus linking integration and differentiation using the naive definition of the integral. This serves to motivate the more robust definition of the Riemann integral in the next section. It will also be a reminder of what you know from calculus and a warmup for what's to come.

Definition 12.1. Let $a < b$, and suppose f is a real-valued function whose domain contains $[a, b]$. Define the **right Riemann Sum** of f on the interval $[a, b]$ with n subintervals to be

$$R_n(f) = \sum_{i=1}^{n} f(x_i)\Delta x \qquad \text{where } \Delta x = \frac{b-a}{n} \text{ and } x_i = a + i\Delta x \tag{12.1}$$

Exercise 12.2. Let $f : [1, 3] \to \mathbb{R}$ be defined by $f(x) = \sqrt{x}$. Use a calculator to find $R_{10}(f)$.

Note that we aren't initially making any assumptions about the function f being "nice" in any way. In particular, f might be badly discontinuous as in the next example.

Exercise 12.3. Let $f : [0,1] \to \mathbb{R}$ be defined by $f(x) = 0$ when x is irrational and $f(x) = 1$ when x is rational. We call this function **Dirichlet's function**. Sketch a graph of f (as best you can), and find a formula for $R_n(f)$.

In calculus, we are taught to think of right Riemann sums as approximations to the exact value of the integral. Visually, for non-negative-valued functions at least, we can represent this by representing the products $f(x_i)\Delta x$ as rectangles; as n tends to ∞, this set of rectangles seems to better and better approximate the region under the curve, as in the demo below that you are likely familiar with.

And thus we make the following definition.

Definition 12.4. Let $a < b$, and suppose f is a real-valued function whose domain contains $[a,b]$. If the sequence $(R_n(f))$ of right Riemann sums of f over $[a,b]$ converges to some real number I, we say that f is **right-integrable** on $[a,b]$. In this case, we write $\mathsf{R}_a^b f$ for I and call this value the **right integral** of f over $[a,b]$.

This definition does two things for us. First it tells us which functions get to be part of our theory—ones where the sequence of right Riemann sums converges. And second it tells us what the value of those integrals should be. While the second part is what we focus on when doing calculations, it's the first part that's important for being able to prove theorems; it lays out the baseline assumptions we can make about the functions being integrated.

The problem with this definition is that those baseline assumptions are too weak to be able to derive theorems from. Definition 12.4 allows too many problematic functions into the theory without being able to properly handle them. The next example hints at the sort of thing that can go wrong.

Exercise 12.5. Show that Dirichlet's function from Exercise 12.3 is right-integrable, and find its right integral. Does the value of the right integral make sense given the sketch you made? Explain.

As one more warmup question to reflect on, the following exercise provides another hint about an arbitrary and possibly-problematic choice we make in the definitions so far.

Exercise 12.6. Why do you think we refer to the expression in Definition 12.1 as the "right" Riemann sum? What would it mean to take the "left" Riemann sum?

12.2 The Domain Additivity Issue

We haven't proven it yet, but it should be true that, for any function that's part of our theory, the integral from a to b plus the integral from b to c of a function should be equal to the integral from a to c. This theorem should make intuitive sense if we are thinking about areas under the curve, but it is also one of the keys to proving the Fundamental Theorem of Calculus.

Unfortunately, right-integrable functions do not necessarily obey the corresponding rule as you'll show below.

Exercise 12.7. For Dirichlet's function f defined in Exercise 12.3, show that f is right-integrable over $\left[0, \frac{1}{\sqrt{2}}\right]$ and over $\left[\frac{1}{\sqrt{2}}, 1\right]$. Find $\mathsf{R}_0^{1/\sqrt{2}} f$ and $\mathsf{R}_{1/\sqrt{2}}^1 f$, and show that

$$\mathsf{R}_0^{1/\sqrt{2}} f + \mathsf{R}_{1/\sqrt{2}}^1 f \neq \mathsf{R}_0^1 f.$$

> **Hint**
>
> In your answer to Exercise 12.7, you can feel free to use familiar facts about whether sums, products, etc. of rational and irrational numbers are rational or irrational without proof, but make sure your facts are actually true!

Because of this and related issues, we are forced to seek an alternative to what we'll call the "naive theory of integration" that arises from Definition 12.4.

To gain a foothold on how to do this, we start by returning to Exercise 12.6, and trying to conceptualize what's going on in the definition of right Riemann sums. When calculating $R_n(f)$, we should imagine dividing the interval $[a, b]$ into n (equal) subintervals, $[x_0, x_1], [x_1, x_2], \ldots, [x_{n-1}, x_n]$. Then each summand $f(x_i)\Delta x$ of Equation 12.1 can be thought of as the width of the ith subinterval (always Δx) times the value of the function at a certain point in that subinterval; namely $f(x_i)$ is the value of the function at the *right* endpoint of the ith interval $[x_{i-1}, x_i]$.

However, the choice to plug in the *right* endpoint of each interval is somewhat arbitrary. For a well-behaved function, we would expect that choosing the left endpoints, midpoints, or any other sample point x_i^* from the ith subinterval would be just as reasonable, and that *as long as the subintervals are all small enough*, adding up the products of the values $f(x_i^*)$ at these sample points with the widths Δx will get us close to the true value of the integral.

Exercise 12.8.

1. Write down a reasonable definition for the **left Riemann sum** $L_n(f)$, what it means for a function to be **left-integrable**, and the **left integral** $\mathsf{L}_a^b f$ of a real-valued function f whose domain includes $[a, b]$.

2. Prove that f is right-integrable on $[a, b]$ if and only if f is left-integrable on $[a, b]$ and that in this case, $\mathsf{R}_a^b f = \mathsf{L}_a^b f$.

Exercise 12.9. For Dirichlet's function f defined in Exercise 12.3, prove that for any $n \in \mathbb{N}$, there exist sample points x_i^* in the ith subinterval such that $\sum_{i=1}^{n} f(x_i^*)\Delta x = 0$.

The previous exercise shows that for Dirichlet's function, the choice of sample points *does* matter. That dependence will be grounds for us to exclude this function from the more rigorous theory that we develop in the remainder of this chapter[1].

Review Questions

1. What does it mean for a function to be "right-integrable"? What is the definition of the "right integral" of such a function?
2. Give an example of a familiar property of integrals that can fail if we adopt the naive theory of integration based on right integrability.

[1]We will return to this idea later, but sophisticated theories of integration that go beyond the scope of this class *do* allow functions like this one and are able to handle them gracefully.

References

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Rock, John A. 2025. *Arbitrarily Close*. 619 Wreath.

Schumacher, Carol. 2008. *Closer and Closer: Introducing Real Analysis*. Jones and Bartlett Publishers.

A Proof-Writing Resources

Below are some proof-writing resources you may find helpful. These are also available as printable cheat sheets: negation.pdf, proof-skeleta.pdf.

A.1 Negating Statements

Often we need to negate logical statements. For example, a problem may ask you prove that a sequence does *not* converge to a, or you may need to set up the contrapositive of an if-then statement.

i "For all" statements

$$\neg\big[(\forall x \in A)P(x)\big] \iff (\exists x \in A)\big[\neg P(x)\big]$$

Example:
Original: "For all integers x, $x^2 > 0$."
Negation: "There exists an integer x such that $x^2 \leq 0$."

i "There exists… such that" statements

$$\neg\big[(\exists x \in A)P(x)\big] \iff (\forall x \in A)\big[\neg P(x)\big]$$

Example:
Original: "There exists $x \geq 0$ such that $\sqrt{x} \notin \mathbb{Q}$."
Negation: "For all $x \geq 0$, $\sqrt{x} \in \mathbb{Q}$."

i Conditionals

$$\neg(P \Rightarrow Q) \iff P \wedge (\neg Q)$$

Remember that many conditionals are often preceded by a hidden "for all'' quantifier.
Example:
Original: "(For all x) if x is even, then x^2 is even."
Negation: "There exists x such that x is even and x^2 is odd."

"And" statements

$$\neg(P \wedge Q) \iff (\neg P) \vee (\neg Q)$$

Remember that many conditionals are often preceded by a hidden "for all'' quantifier.
Example:
Original: "There exists $n \in \mathbb{N}$ such that n divides 15 and 2 divides n."
Negation: "For all $n \in \mathbb{N}$, either n does not divide 15 or 2 does not divide n."

"Or" statements

$$\neg(P \vee Q) \iff (\neg P) \wedge (\neg Q)$$

Remember that many conditionals are often preceded by a hidden "for all'' quantifier.
Example:
Original: "For all $n \in \mathbb{Z}$, n is even or n is odd."
Negation: "There exists $n \in \mathbb{Z}$ such that, n is not even and n is not odd."

A.2 Proof skeleta

Below are what I call "proof skeleta"—basic structures you can use when implementing standard proof techniques.

Conditional Statements (Direct)

Goal: For propositions P, Q, proving $P \Rightarrow Q$.
Proof. [State any upfront assumptions.] Assume P. *[Use definitions and known results to derive Q.]* Therefore Q.

Conditional Statements (Contrapositive)

Goal: For propositions P, Q, proving $P \Rightarrow Q$.
Proof. [State any upfront assumptions.] Assume $\neg Q$. *[Use definitions and known results to derive $\neg P$.]* Therefore $\neg P$.

Biconditionals

Goal: For propositions P, Q, proving $P \Leftrightarrow Q$.
Proof. [State any upfront assumptions.] We will prove implication both ways.

- ($\Rightarrow$) *[Prove $P \Rightarrow Q$.]*
- ($\Leftarrow$) *[Prove $Q \Rightarrow P$.]*

Therefore P if and only if Q.

i "The following are equivalent"

Goal: Proving propositions P, Q, and R are logically equivalent.
Proof. [State any upfront assumptions.] We will prove $P \Rightarrow Q$, $Q \Rightarrow R$, and $R \Rightarrow P$.

- ($P \Rightarrow Q$) *[Prove $P \Rightarrow Q$.]*
- ($Q \Rightarrow R$) *[Prove $Q \Rightarrow R$.]*
- ($R \Rightarrow P$) *[Prove $R \Rightarrow P$.]*

Therefore P, Q, and R are logically equivalent.

- The order of P, Q, R is unimportant, and the same idea can be applied to more than three propositions.

i Subsets

Goal: For sets A, B, proving $A \subseteq B$.
Proof. [State any upfront assumptions.] Let $a \in A$. *[Use definitions and known results to deduce $a \in B$.]* Therefore $a \in B$.

i Set Equality

Goal: For sets A, B, proving $A = B$.
Proof. [State any upfront assumptions.] We will prove inclusion both ways.

- ($\subseteq$) *[Prove $A \subseteq B$.]*
- ($\supseteq$) *[Prove $B \subseteq A$.]*

Therefore $A = B$.

i "For all" statements

Goal: Proving $(\forall x \in A)P(x)$.
Proof. [State any upfront assumptions.] Suppose $x \in A$. *[Use definitions and known results to deduce $P(x)$.]* Therefore for all $x \in A$, $P(x)$.

i "There exists… such that" statements

Goal: Proving $(\exists x \in A)P(x)$.
Proof. [State any upfront assumptions.]

- *[Exhibit a candidate $x \in A$.]*
- *[Use definitions and known results to deduce $P(x)$.]*

Therefore there exists $x \in A$ such that $P(x)$.

i Uniqueness

Goal: Proving $(\exists! x \in A)P(x)$.
Proof. [State any upfront assumptions.]

- *[Prove $(\exists x \in A)P(x)$.]*
- Suppose $x_1, x_2 \in A$ and $P(x_1)$ and $P(x_2)$ are true. *[Use definitions and known results to deduce $x_1 = x_2$.]* Therefore $x_1 = x_2$.

Therefore there exists a unique $x \in A$ such that $P(x)$.

i Proof by Contradiction

Goal: Proving a proposition P.
Proof. [State any upfront assumptions.] By way of contradiction, assume $\neg P$. *[Use definitions and known results to arrive at a contradiction.]* Therefore P.

- P may be a conditional statement, but it doesn't have to be.
- The contradiction you arrive at may be $(\neg P) \wedge P$, or it may be something else.

i Proof by Induction

Goal: Proving $(\forall n \in \mathbb{N})P(n)$.
Proof. [State any upfront assumptions.] We will utilize a proof by induction.

- **Base case.** *[Verify $P(1)$ is true.]* Therefore $P(1)$.
- **Inductive step.** Let $k \in \mathbb{N}$. Assume $P(k)$. *[Use definitions and known results to deduce $P(k+1)$.]*

Therefore $P(n)$ is true for all $n \in \mathbb{N}$.

Function Equality

Goal: For two functions $f, g : A \to B$, prove $f = g$.
Proof. [State any upfront assumptions.] Let $x \in A$. *[Use definitions and known results to deduce $f(x) = g(x)$.]* Therefore $f(x) = g(x)$ for all $x \in A$, so $f = g$.

Injectivity

Goal: Prove that $f : A \to B$ is injective.
Proof. [State any upfront assumptions.] Let $x_1, x_2 \in A$ and assume $f(x_1) = f(x_2)$. *[Use definitions and known results to deduce $x_1 = x_2$.]* Therefore $x_1 = x_2$ whenever $f(x_1) = f(x_2)$, so f is injective.

Surjectivity

Goal: Prove that $f : A \to B$ is surjective.
Proof. [State any upfront assumptions.] Let $y \in B$. *[Use definitions and known results to prove $(\exists x \in A)(f(x) = y)$.]* Therefore, for all $y \in B$, there exists $x \in A$ such that $f(x) = y$, so f is surjective.

Bijectivity

Goal: Prove that $f : A \to B$ is bijective.
Proof. [State any upfront assumptions.]

- *[Prove f is injective.]*
- *[Prove f is surjective.]*

Therefore f is both injective and surjective, so f is bijective.

B Collections of Sets

In set theory, we often need to think about not just one set, but a whole *collection*[1] of sets.

Some examples that appear in the text are below.

Example B.1. One distinguishing feature of the real numbers is, intuitively, if you keep zooming in on the number line, you are always zooming in *to a number*. For example, consider the following collection of intervals:

$$\begin{aligned} U_1 &= [3, 4] \\ U_2 &= [3.1, 3.2] \\ U_3 &= [3.14, 3.15] \\ U_4 &= [3.141, 3.142] \\ U_5 &= [3.1415, 3.1416] \\ U_6 &= [3.14159, 3.14160] \\ &\vdots \end{aligned}$$

If the universe is $\mathbb{R}$, there is a number that will be in U_i for all i, the number π. By contrast, in $\mathbb{Q}$, there is no number inside all of these sets—we'd be zooming in onto something that isn't there.

Example B.2. In a metric space, an open set S is a union of *a collection of* open balls.

Often we think about putting an open ball around every point of S, each with a radius small enough so that the ball is a subset of S.

Example B.3. To say that a subset of a metric space is *unbounded* means that it does not fit in any open ball. One technique for proving that a set V is unbounded is to pick any point and consider the collection of balls of radii 1,2,3,… around that point and show that V contains a point in the complement of each ball.

The examples above show some of the operations we might consider performing when we have a collection of sets—unions, intersections, and complements. We collect some basic facts and notation below.

[1] Really this is the same as a *set of sets*, but we use the word "collection" to help distinguish between the two levels of the hierarchy.

B.1 Indexing Sets

Typically when we have a collection of sets, we want to put those sets in one-to-one correspondence with some set called the *indexing set*. For example, if we want one set in our collection for each integer, our indexing set would be $\mathbb{Z}$, and we might write U_k for the sets in our collection where $k \in \mathbb{Z}$; so there would be a U_7 and a U_0 and a U_{-3}, for instance. For the entire collection, we would write $\{U_k\}_{k\in\mathbb{Z}}$.

More generally, if our sets are indexed by Δ, we would write U_α for a set in our collection with $\alpha \in \Delta$, and we would write $\{U_\alpha\}_{\alpha\in\Delta}$ for the collection.

When we have finitely many sets, it is typical to call them something like $U_1, U_2, \ldots, U_n$; the collection would then be the set of sets $\{U_1, U_2, \ldots, U_n\}$. In this case $\Delta = \{1, 2, \ldots, n\}$ is the indexing set since there is one set in our collection for each element of Δ.

Often, we want one set for each natural number. In this case, we can use an indexing set of $\Delta = \mathbb{N}$ and we once again call our sets $U_1, U_2, \ldots$. To indicate the collection of sets, we can write $\{U_i\}_{i\in\mathbb{N}}$ or more commonly $\{U_i\}_{i=1}^{\infty}$.

Occasionally we want to put our sets in correspondence with some other set Δ. In Example B.2, for example, we wanted one set (one open ball) for each point of S. In this case we would make $\Delta = S$. Then for each point $x \in S$, there is a corresponding open ball B_x centered around x. In this case, we would write $\{B_x\}_{x\in S}$ for the collection.

B.2 Unions and Intersections

For two sets U and V, we are familiar with the idea that

- The *union* $U \cup V$ consists of all elements that are in *at least one of* U or V.
- The *intersection* $U \cap V$ consists of all elements are in *both* U and V.

Extending this to arbitrary collections of sets, if $\{U_\alpha\}_{\alpha\in\Delta}$ is a collection of sets, we write

$$\bigcup_{\alpha\in\Delta} U_\alpha = \{x \ : \ x \in U_\alpha \text{ for at least one } \alpha \in \Delta\}$$
$$\bigcap_{\alpha\in\Delta} U_\alpha = \{x \ : \ x \in U_\alpha \text{ for all } \alpha \in \Delta\}$$

In the common special case of where $\Delta = \mathbb{N}$, we instead write $\bigcup_{i=1}^{\infty} U_i$ and $\bigcap_{i=1}^{\infty} U_i$.

Relating this back to our examples, in Example B.1, we are claiming that in the real numbers, $\bigcap_{i=1}^{\infty} U_i$ would be $\{\pi\}$ since π and no other number is in *all* of these intervals, but if our universe were $\mathbb{Q}$, this intersection would be $\emptyset$ since *no* rational number lives in all of these intervals.

In Example B.2, we are saying that S is an open set if there exists a collection of open balls $\{B_\alpha\}$ such that $\bigcup B_\alpha = S$.[2] And similarly, if S is open, for each $x \in S$, there exists an open ball B_x centered at x such that $\bigcup_{x \in S} B_x = S$. If you can wrap your head around that, you are in good shape!

B.3 DeMorgan's Laws

Recall that the **complement** of a set S, for which we'll write S^{C}, is all of the elements of our understood universe that are not in S.

DeMorgan's Laws tell us how complementation interacts with unions and intersections. For two sets, DeMorgan's Laws tell us

$$(U \cup V)^{\mathsf{C}} = U^{\mathsf{C}} \cap V^{\mathsf{C}}$$
$$(U \cap V)^{\mathsf{C}} = U^{\mathsf{C}} \cup V^{\mathsf{C}}$$

But these same basic rules apply to arbitrary collections of sets as well.

Theorem B.4 (DeMorgan's Laws)**.** *For any collection of sets* $\{U_\alpha\}_{\alpha \in \Delta}$*, we have*

$$\left(\bigcup_{\alpha \in \Delta} U_\alpha \right)^{\mathsf{C}} = \bigcap_{\alpha \in \Delta} U_\alpha^{\mathsf{C}}, \text{ and}$$
$$\left(\bigcap_{\alpha \in \Delta} U_\alpha \right)^{\mathsf{C}} = \bigcup_{\alpha \in \Delta} U_\alpha^{\mathsf{C}}.$$

Proof. For the first equation, note that $x \in \left(\bigcup_{\alpha \in \Delta} U_\alpha\right)^{\mathsf{C}}$ means that x is not in $\bigcup_{\alpha \in \Delta} U_\alpha$, which is equivalent to saying x is not in any of the U_α. This is in turn equivalent to saying $x \in U_\alpha^{\mathsf{C}}$ for all $\alpha \in \Delta$, meaning $x \in \bigcap_{\alpha \in \Delta} U_\alpha^{\mathsf{C}}$. □

Exercise B.5. Prove that second equation from DeMorgan's Laws is true.

[2] Note that I am using shorthand here by omitting the indexing set since it isn't important in this case.

Hint

There are at least two ways to do this. First is to give an argument similar to the proof of the first equation. The other is to *apply* the first equation, which is already proven to see that the second one is true.

C The Cauchy-Schwarz Inequality

In Chapter 5, we defined the *standard distance function* on $\mathbb{R}^n$ to be

$$d(\mathbf{a}, \mathbf{b}) = \sqrt{(b_1 - a_1)^2 + \cdots + (b_n - a_n)^2}$$

where $\mathbf{a} = (a_1, \ldots, a_n)$, $\mathbf{b} = (b_1, \ldots, b_n)$.

we claimed that the standard distance function on $\mathbb{R}^n$ satisfies the Triangle Inequality, but we never proved it. That is, for any $\mathbf{a}, \mathbf{b}, \mathbf{c} \in \mathbb{R}^n$, we have

$$d(\mathbf{a}, \mathbf{c}) \leq d(\mathbf{a}, \mathbf{b}) + d(\mathbf{b}, \mathbf{c}). \tag{C.1}$$

As a first step, we will prove the Cauchy-Schwarz Inequality.

Theorem C.1 (Cauchy-Schwarz Inequality)**.** *Suppose $u_1, \ldots, u_n \in \mathbb{R}$ and $v_1, \ldots, v_n \in \mathbb{R}$. Then*

$$\left(\sum_{i=1}^{n} u_i v_i\right)^2 \leq \left(\sum_{i=1}^{n} u_i^2\right)\left(\sum_{i=1}^{n} v_i^2\right).$$

Proof. Define a polynomial

$$p(t) = \sum_{i=1}^{n} (t u_i + v_i)^2.$$

Note that $p(t)$ is never negative. Its formula can be rewritten as

$$p(t) = \sum_{i=1}^{n} t^2 u_i^2 + 2 t u_i v_i + v_i^2 = t^2 \left(\sum_{i=1}^{n} u_i^2\right) + 2t \left(\sum_{i=1}^{n} u_i v_i\right) + \left(\sum_{i=1}^{n} v_i^2\right).$$

Note that $p(t)$ is a quadratic function, so since it is never negative, its discriminant Δ is non-positive. That is

$$0 \geq \Delta = \left(2 \sum_{i=1}^{n} u_i v_i\right)^2 - 4 \left(\sum_{i=1}^{n} u_i^2\right)\left(\sum_{i=1}^{n} v_i^2\right)$$

which implies

$$\left(\sum_{i=1}^{n} u_i^2\right)\left(\sum_{i=1}^{n} v_i^2\right) \geq \left(\sum_{i=1}^{n} u_i v_i\right)^2.$$

□

Now that we have the Cauchy-Schwarz Inequality, the rest of the proof is just a matter of applying it and doing some algebra.

Corollary C.2. *The standard distance function on $\mathbb{R}^n$ satisfies the Triangle Inequality.*

Proof. Squaring both sides of Equation C.1, we will prove that for points $\mathbf{a} = (a_1, \ldots, a_n)$, $\mathbf{b} = (b_1, \ldots, b_n)$, and $\mathbf{c} = (c_1, \ldots, c_n)$,

$$d(\mathbf{a}, \mathbf{c})^2 \leq (d(\mathbf{a}, \mathbf{b}) + d(\mathbf{b}, \mathbf{c}))^2.$$

Expanding the right side, we have

$$(d(\mathbf{a}, \mathbf{b}) + d(\mathbf{b}, \mathbf{c}))^2 = d(\mathbf{a}, \mathbf{b})^2 + 2d(\mathbf{a}, \mathbf{b})d(\mathbf{b}, \mathbf{c}) + d(\mathbf{b}, \mathbf{c})^2.$$

Letting $u_i = b_i - a_i$ and $v_i = c_i - b_i$, this becomes

$$\left(\sum_{i=1}^{n} u_i^2\right) + 2\sqrt{\left(\sum_{i=1}^{n} u_i^2\right)\left(\sum_{i=1}^{n} v_i^2\right)} + \left(\sum_{i=1}^{n} v_i^2\right),$$

which, by Theorem C.1, is at least

$$\left(\sum_{i=1}^{n} u_i^2\right) + 2\left(\sum_{i=1}^{n} u_i v_i\right) + \left(\sum_{i=1}^{n} v_i^2\right).$$

However, this last line is equal to

$$\sum_{i=1}^{n} (u_i + v_i)^2 = d(\mathbf{a}, \mathbf{c})^2,$$

and so we have proven the theorem. □

D The Well-Ordering Property

One of the basic assumptions we make about the number system $\mathbb{N}$ that underwrites our ability to use mathematical induction is what's known as the Well-Ordered Axiom.

Axiom D.1 (The Well-Ordering Axiom)**.** *Let S be any nonempty subset of $\mathbb{N}$. Then S has a least element.*

We treat this as an axiom, which means we assume it without proof. While not true as written for $\mathbb{Z}$, we can adapt it to integers with an additional assumption.

Theorem D.2 (The Well-Ordering Property of $\mathbb{Z}$)**.** *Let S be any nonempty subset of $\mathbb{Z}$ that is bounded below by an element of $\mathbb{Z}$. Then S has a least element.*

Proof. Suppose $n \in \mathbb{Z}$ is a lower bound on S. Then define

$$T = \{1 - n + s \ : \ s \in S\}$$

Then if $t \in T$, $t = 1 - n + s$ for some $s \in S$,

$$t = 1 - n + s \geq 1 - n + n = 1.$$

Thus all elements of T are positive integers, so $T \subseteq \mathbb{N}$. So by Axiom D.1, T has a least element m; that is $m \in T$ and $m \leq t$ for all $t \in T$. Thus $m = 1 - n + k$ for some $k \in S$, so $m + n - 1 = k \in S$. And if $s \in S$, then $1 - n + s \in T$, so $m \leq 1 - n + s$, and therefore $k = m + n - 1 \leq s$. Thus k is the least element of S. □

Exercise D.3. Would Theorem D.2 be true if we replaced $\mathbb{Z}$ with $\mathbb{Q}$? or $\mathbb{R}$? Give a proof or counterexample for each.

E Links to Desmos Web Apps

Below are links to the web app versions of the Desmos interactives found throughout the textbook.

E.1 Axioms of $\mathbb{R}$

1. Supremum of $[0, 3)$
2. Nested Interval Theorem
3. Uncountability of $\mathbb{R}$ using NIT

E.2 Metric Spaces

1. Distance on the number line
2. Distance between functions
3. Suprema and distance
4. Open set definition demo
5. Open set as a union of open balls
6. Open half plane as a union of open balls
7. Boundedness definition
8. Boundedness in $\mathbb{R}$